

# The Projection Spectral Layer Theory (PSLT): A Rank-2 Computable Closure for the Three-Generation Structure and Higgs Signal Strength

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We present a comprehensive, computable EFT-level “closed chain” for the Projection Spectral Layer Theory (PSLT) that naturally generates a self-consistent three-generation structure of matter. The framework is built on a minimal Einstein–Yang–Mills–Higgs (EYMH) effective action with non-minimal curvature coupling, where the background geometry induces a layer-indexed spectral scale  $\omega_N$ . The theoretical closure is achieved by unifying three key modules: (i) a **Micro-degeneracy module** ( $g_N$ ) implemented in baseline `fp_2d_full` mode (bounded microcanonical low- $N$  window + energy-gap tail prescription, phase-space normalized at  $N = 3$ ); (ii) a **Rank-2 Computable Kinetic module** ( $\Gamma_N$ ) defined as the largest positive eigenvalue of a  $2 \times 2$  growth matrix with explicit WKB tunneling suppression; and (iii) a **Visibility module** ( $B_N$ ) implemented from overlap-defined  $y_N^{\text{eff}}(D)$  profiles extracted by the same 2D localized operator chain, with branch tracking and microcanonical window averaging for mode-switch control, and high- $N$  saturation ( $B_{N>3} = 1$ ) to avoid double-counting with the  $g_N$  regulator. The resulting normalized layer probability  $P_N(t_{\text{coh}})$  demonstrates a stable “winner” phase diagram with clear dominance of layers  $N = 1, 2, 3$  over a significant region of the geometric parameter space  $(D, \eta)$ . We verify the closure against two benchmarks: (1) internal stability of the three-generation hierarchy (Generation Ratio  $> 90\%$  over 92.7% of the sampled  $(D, \eta)$  grid), and (2) a scan-level EFT/Wilson-matched compatibility test with the ATLAS Run-3  $H \rightarrow \mu\mu$  signal strength ( $\mu_{\mu\mu}^{\text{obs}} = 1.4 \pm 0.4$ ). The EFT/Wilson-matched mapping yields a restricted acceptance band (about 16.1% of the scanned grid for  $\chi^2 < 4$ ), providing a sharper falsifiable constraint on  $(D, \eta)$ .

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## I. INTRODUCTION

The existence of exactly three generations of fermions is one of the most persistent puzzles in the Standard Model (SM) of particle physics. While the SM accommodates three families through the CKM and PMNS mixing matrices, it offers no dynamical reason for why the number is three, nor why the mass hierarchy spans such a vast range. Traditional attempts to explain this structure often rely on complex discrete symmetries or string logic compactifications that, while elegant, can be difficult to falsify directly.

The Projection Spectral Layer Theory (PSLT) proposes a different paradigm: *Generation structure is a spectral consequence of the underlying projection geometry of spacetime.* In this framework, "generations" are not ad-hoc copies but rather distinct *spectral layers* ( $N = 1, 2, 3, \dots$ ) of vacuum excitations, selected by a competition between microstate degeneracy (entropy) and geometric formation rates (kinetics). This structural viewpoint is complementary to recent Clifford-algebra work where three families emerge from an intrinsic  $S_3$  organization rather than manual generation copying [1]; the emphasis here is different: a map-level computable closure with explicit scan-level falsifiability.

In this work, we consolidate the complete theoretical framework of PSLT into a single, falsifiable "closed chain". We upgrade earlier effective descriptions to a **computable** rank-2 dynamical system with an action-derived extraction subchain and an action-informed global scan chain, where  $\chi_{LR}(D)$  and  $\tilde{A}_\ell(D)$  are now injected on the full scan grid from direct action-derived profiles. The core deliverable is a map from geometric control parameters  $(D, \eta)$ —representing dual-center separation and overlap—to an experimentally observable layer probability  $P_N$ .

Our primary contributions are:

1. **Theoretical Motivation:** We present three convergent arguments for the micro-degeneracy  $g_N$  (Route A: Mock-modular, Route B: ER=EPR, Route C: QG Template), and implement a scan-ready baseline through the bounded `fp_2d_full` profile with controlled high- $N$  tail prescription.
2. **Rank-2 Computable Kinetics:** We replace heuristic growth rates with a rigorous eigenvalue problem for the formation matrix  $\mathbf{M}_N$ , incorporating numerical WKB tunneling suppression.
3. **Overlap-Defined Visibility:** We promote  $B_N$  to overlap-defined  $y_N^{\text{eff}}(D)$  profiles from the 2D localized solver, stabilized by branch tracking and microcanonical windows; a Yukawa law is retained only as a legacy comparator/fallback.

4. **Verification:** We demonstrate stable three-generation dominance at  $\mathcal{R}_3 > 90\%$  and a constrained EFT/Wilson-matched-compatible band for LHC  $H \rightarrow \mu\mu$ .

## II. THEORETICAL FRAMEWORK

### A. The Closed Chain Equation

The central output of the PSLT framework is the normalized layer occupancy probability  $P_N(t_{\text{coh}})$ , defined by the competition between entropic weight ( $W_{\text{entropy}}$ ) and kinetic formation ( $W_{\text{kinetic}}$ ). The master equation is:

$$P_N(t_{\text{coh}}; D, \eta) = \frac{W_N(t_{\text{coh}})}{\sum_K W_K(t_{\text{coh}})}, \quad W_N = B_N \cdot g_N \cdot \left(1 - e^{-\Gamma_N(D, \eta)t_{\text{coh}}}\right). \quad (1)$$

This equation unifies the three critical modules:

- $g_N$ : **Micro-degeneracy** (Entropy). The number of available microstates in layer  $N$ .
- $\Gamma_N$ : **Dynamical Rate** (Kinetics). The rate at which these states can form/tunnel from the vacuum.
- $B_N$ : **Visibility Factor** (Observation). The coupling strength of layer  $N$  to the observable sector.

Figure 1 illustrates the logical flow of the PSLT framework.

### B. Action-Derived Chain at a Glance

For implementation-level clarity, Figure 2 summarizes the action-derived operator chain and its interfaces to scan-level diagnostics in one page. The executable, clone-level Standard Operating Procedure (SOP) is provided in Appendix A.

### C. Geometric Foundation: Minimal EYMH + Curvature

We start from a minimal Einstein–Yang–Mills–Higgs (EYMH) effective action supplemented by a non-minimal curvature coupling  $\xi R|\Phi|^2$ :

$$S = \int d^4x \sqrt{-g} \left[ \frac{M_{\text{Pl}}^2}{2} R - \frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} - (D_\mu \Phi)^\dagger (D^\mu \Phi) - \lambda(|\Phi|^2 - v^2)^2 - \xi R|\Phi|^2 \right]. \quad (2)$$

PSLT posits that the background geometry is a *projection* from a higher-dimensional manifold, inducing a locally conformally flat metric:

$$g_{\mu\nu}(x) = \Omega^2(x) \eta_{\mu\nu}. \quad (3)$$

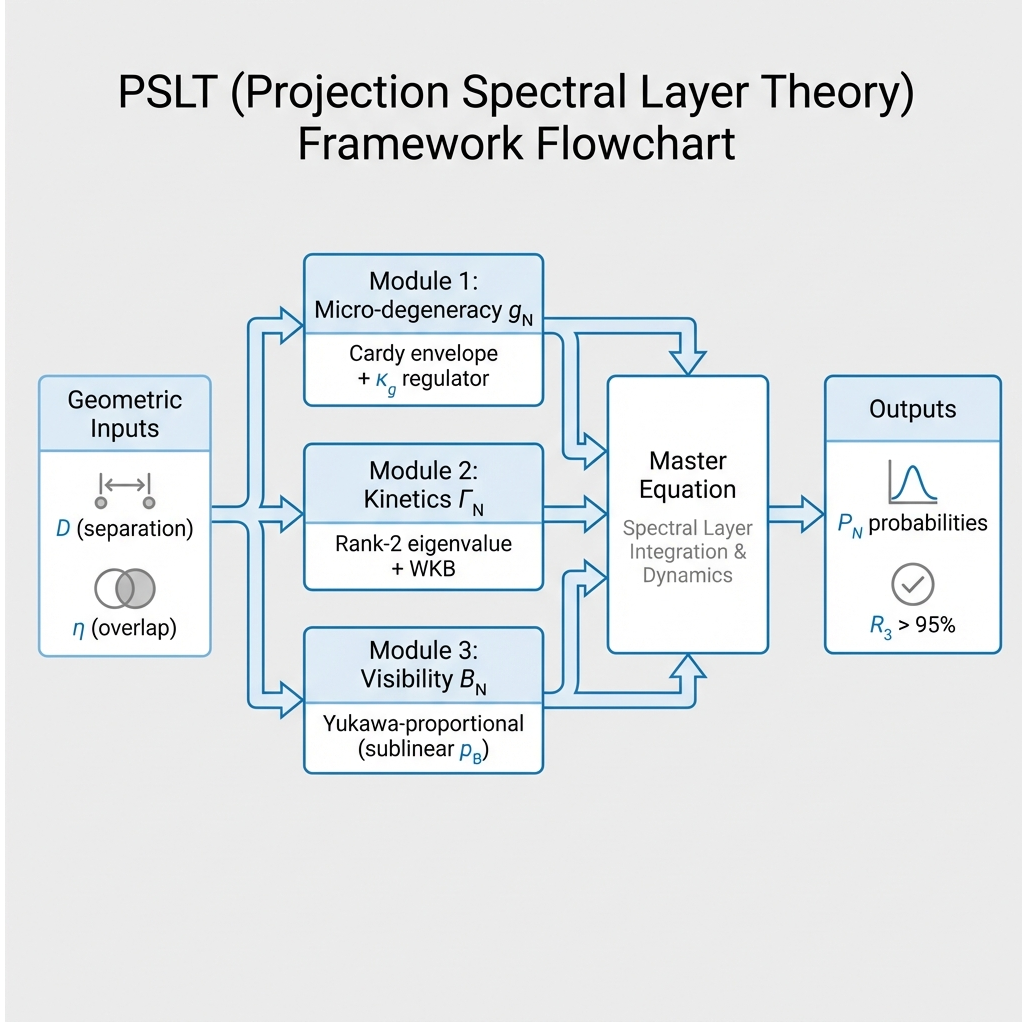


FIG. 1. Schematic overview of the PSLT framework. Geometric inputs ( $D, \eta$ ) feed into three modules (micro-degeneracy  $g_N$ , kinetic rate  $\Gamma_N$ , visibility  $B_N$ ), which combine in the master equation to yield the observable layer probabilities  $P_N$  and the three-generation ratio  $R_3$ . In the baseline chain, high- $N$  control is implemented by the bounded phase-space-tail prescription in `fp_2d_full`, while  $B_{N>3}$  saturates to unity.

Under conformal rescaling  $\Phi \rightarrow \Omega^{-1}\tilde{\Phi}$ , the scalar equation of motion reduces to a Schrödinger-like eigenproblem:

$$[-\nabla^2 + V_{\text{eff}}(x)]\psi_N = \omega_N^2 \psi_N, \quad (4)$$

where the effective potential  $V_{\text{eff}}$  is determined by the conformal factor  $\Omega(x)$  and the specific projection geometry (e.g., stereographic projection of dual centers).

The layer index  $N$  emerges naturally as the principal quantum number of this spectrum. In Section III, we derive the spectral scale  $\mu(D)$  explicitly from a two-center geometry; the result supports a power-law scaling  $\mu(D) \propto D^{-\gamma}$  with  $\gamma \approx 0.1$  in the demonstrator regime. For

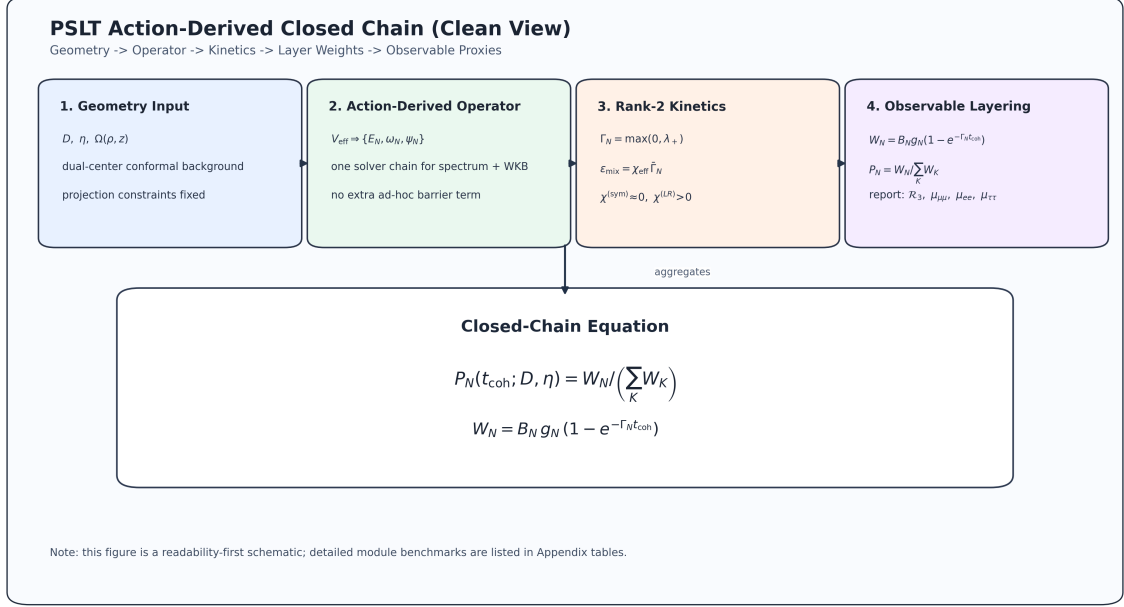


FIG. 2. One-page action-derived chain overview used in this manuscript: geometry input  $\rightarrow$  operator extraction  $\rightarrow$  scan diagnostics  $\rightarrow$  observable EFT/Wilson checks. This diagram is aligned with the repository reproducibility entrypoint and Appendix A.

analytic tractability, we adopt the **hydrogenic proxy**:

$$\omega_N(D) \simeq \mu(D) \left( 1 - \frac{1}{2N^2} \right), \quad \mu(D) = \mu_0 D^{-\gamma}, \quad (5)$$

where  $\mu_0$  and  $\gamma$  are determined numerically from the geometry (Section III). This form captures the essential  $N$ -dependence from the hydrogenic spectrum while allowing the  $D$ -dependence to be set by explicit calculation. For transparency, we additionally provide a bound-state exact benchmark for  $(\omega_1, \omega_2)$  at  $D = \{6, 12, 18\}$  with coarse/mid/fine convergence from the same action-derived operator chain (Appendix B8, Table VIII). These benchmark values use  $E = \omega^2 - m_0^2 < 0$  and should not be mixed with the generalized localized-extraction eigenvalues  $\lambda$  in Appendix D.

#### D. Model-Chain Assumption Status

For transparency, Table I separates action-derived components from EFT-level surrogates currently used in the global scan.

TABLE I. Status map of the PSLT closed chain in the present manuscript.

| Component                                                                 | Current Status                                                                                                                         | Used in Global Scan            | Next Upgrade Path                                                                                                         |
|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| $\Omega(\rho, z; D)$ geometry                                             | two-center Green-function solution of projected Poisson constraint (two-source model)                                                  | Yes                            | Derive the projected source term from a specific higher-dimensional parent solution                                       |
| $V_{\text{eff}}$ , low modes, WKB action                                  | Action-derived + numerically converged                                                                                                 | Yes                            | Extend same solver chain to full $(D, \eta, N)$ map                                                                       |
| $g_N$ phase-space full-tail mode (fp.2d.full)                             | Baseline mode + legacy Cardy reference retained with phase-space normalization                                                         | Yes                            | Promote full $(D, \eta, N)$ action-derived extraction beyond map-level profile closure                                    |
| $\chi_N^{(LR)}(D)$ channel                                                | localized Stabilized direct full-scan profile (Dgrid60) 2D fine extraction + scan-level profile injection (Appendix D; D60 bias audit) | Yes                            | Promote to full per- $(D, \eta, N)$ localized direct projection (no profile injection)                                    |
| $\Gamma_{N,\ell}$ channel normalization ( $A_\ell$ )                      | Action-derived baseline with stabilized localized-direct Dgrid60 profile support and legacy constant- $A_\ell$ comparator              | Yes                            | Extend to fully localized channel-resolved direct extraction over $(D, \eta, N)$                                          |
| Open-system (Lindblad)                                                    | $\chi_{\text{eff}}$ Scan-ready diagnostic module with profiled $\gamma_\phi(D), \gamma_{\text{mix}}(D)$ (Appendix E 4)                 | Diagnostic only (not baseline) | Derive microscopic EYMH bath $(L_k, \gamma_k)$ and then promote to baseline if robust                                     |
| $B_N$ visibility profile $B_N(D)$ from overlap-defined $y_N^{\text{eff}}$ | Action-derived profile extraction + tracked microcanonical windowing                                                                   | Yes                            | Upgrade to fully normalized EFT operator matching (Section VIB)                                                           |
| $H \rightarrow \mu\mu$ mapping [Eq. (51)]                                 | EFT/Wilson-matched baseline + UV-tree/LL-RG audit map                                                                                  | Yes                            | Extend current UV-tree + LL-RG closure to full UV-to-EFT matching from EYMH action (operator basis + loop-level matching) |



### III. GEOMETRY-TO-SPECTRUM WORKED EXAMPLE

This section demonstrates the complete derivation chain  $\Omega(x; D) \rightarrow V_{\text{eff}}(x) \rightarrow \omega_N(D)$ , establishing the geometry-to-spectrum correspondence as a computational reality rather than an ansatz.

#### A. Two-Center Harmonic Conformal Factor

The geometric origin of the dual-center conformal factor is formulated as a projected Poisson constraint on the static spatial slice:

$$\nabla^2 \Omega(\mathbf{x}) = -4\pi \sigma(\mathbf{x}), \quad \sigma(\mathbf{x}) \equiv a[\rho_\varepsilon(\mathbf{x} - \mathbf{x}_+) + \rho_\varepsilon(\mathbf{x} - \mathbf{x}_-)]. \quad (6)$$

Here  $\mathbf{x}_\pm = \pm(D/2)\hat{z}$  are the two projected centers and  $\rho_\varepsilon$  is a normalized smeared source. With Green's function  $G(\mathbf{x}) = (4\pi|\mathbf{x}|)^{-1}$  for  $-\nabla^2$ , the unique asymptotically flat solution ( $\Omega \rightarrow 1$  at  $|\mathbf{x}| \rightarrow \infty$ ) is

$$\Omega(\mathbf{x}) = 1 + \int d^3x' \frac{\sigma(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}. \quad (7)$$

Choosing the Plummer-type kernel

$$\rho_\varepsilon(\mathbf{r}) = \frac{3\varepsilon^2}{4\pi(|\mathbf{r}|^2 + \varepsilon^2)^{5/2}}, \quad \int d^3r \rho_\varepsilon(\mathbf{r}) = 1, \quad (8)$$

one obtains the closed form

$$\boxed{\Omega(\rho, z; D) = 1 + a \left( \frac{1}{r_+} + \frac{1}{r_-} \right), \quad r_\pm = \sqrt{\rho^2 + (z \mp D/2)^2 + \varepsilon^2}} \quad (9)$$

in cylindrical coordinates  $(\rho, z)$  (axisymmetric,  $m = 0$ ). The corresponding source identity follows from

$$\nabla^2 \left( \frac{1}{\sqrt{r^2 + \varepsilon^2}} \right) = -\frac{3\varepsilon^2}{(r^2 + \varepsilon^2)^{5/2}}, \quad (10)$$

hence

$$\nabla^2 \Omega = -4\pi a [\rho_\varepsilon(x - x_+) + \rho_\varepsilon(x - x_-)], \quad \rho_\varepsilon(x) = \frac{3\varepsilon^2}{4\pi(|x|^2 + \varepsilon^2)^{5/2}}. \quad (11)$$

Equation (9) therefore is not an arbitrary fitting form: once Eq. (6), two-center source support, and asymptotic flatness are fixed, it is the corresponding Green-function solution.

## B. Action-Derived Effective Potential

We derive  $V_{\text{eff}}$  directly from the Klein-Gordon equation in curved spacetime. Consider a scalar field with non-minimal curvature coupling:

$$(\square_g - m_0^2 - \xi R)\Phi = 0. \quad (12)$$

For time-harmonic modes  $\Phi = e^{-i\omega t}\phi(\mathbf{x})$  in the background  $g_{\mu\nu} = \Omega^2\eta_{\mu\nu}$ , we perform the conformal field rescaling  $\phi = \Omega^{-1}\psi$ . After explicit calculation of  $\square_g$  and cancellation of first-derivative terms (see Appendix B 2), the equation reduces to:

$$\boxed{[-\nabla^2 + V_{\text{eff}}(\mathbf{x})]\psi = \omega^2\psi} \quad (13)$$

with the **action-derived** effective potential:

$$\boxed{V_{\text{eff}} = m_0^2\Omega^2 + (1 - 6\xi)\Omega^{-1}\nabla^2\Omega} \quad (14)$$

This is the **only form** consistent with the action—no engineered barrier or box terms appear.

*a. Key properties.*

1. At  $\xi = \xi_c = 1/6$  (conformal coupling in 4D), the derivative term vanishes:  $V_{\text{eff}} \rightarrow m_0^2\Omega^2$ .
2. As  $r \rightarrow \infty$ :  $\Omega \rightarrow 1$ ,  $\nabla^2\Omega \rightarrow 0$ , so  $V_{\text{eff}} \rightarrow m_0^2$  (continuum threshold).
3. Near each center:  $\nabla^2\Omega < 0$  (smeared source), so for  $\xi < 1/6$  the potential develops attractive wells.

We define the shifted potential  $U \equiv V_{\text{eff}} - m_0^2$  so that  $U \rightarrow 0$  at infinity. Bound states satisfy  $E = \omega^2 - m_0^2 < 0$ . Figure 3 shows the decomposition.

## C. Numerical Solution

We solve the 2D axisymmetric eigenproblem using finite differences on a  $(\rho, z)$  grid with Eq. (14):

$$\left[ -\frac{\partial^2}{\partial \rho^2} - \frac{1}{\rho} \frac{\partial}{\partial \rho} - \frac{\partial^2}{\partial z^2} + V_{\text{eff}}(\rho, z; D) \right] \psi_N = \omega_N^2 \psi_N. \quad (15)$$

Boundary conditions:  $\partial_\rho \psi|_{\rho=0} = 0$  (axis regularity),  $\psi|_{\text{boundary}} = 0$  (Dirichlet). Parameters:  $a = 1.0$ ,  $\varepsilon = 0.2$ ,  $m_0 = 1.0$ ,  $\xi = 0.0$ . Grid:  $(n_\rho, n_z) = (50, 500)$  with fine resolution  $\Delta z \ll \varepsilon$  near the cores.

**Critical numerical requirement:** The grid spacing must satisfy  $\Delta z \ll \varepsilon$  to resolve the smeared source structure. This is essential for capturing the deep negative wells in  $U$ .

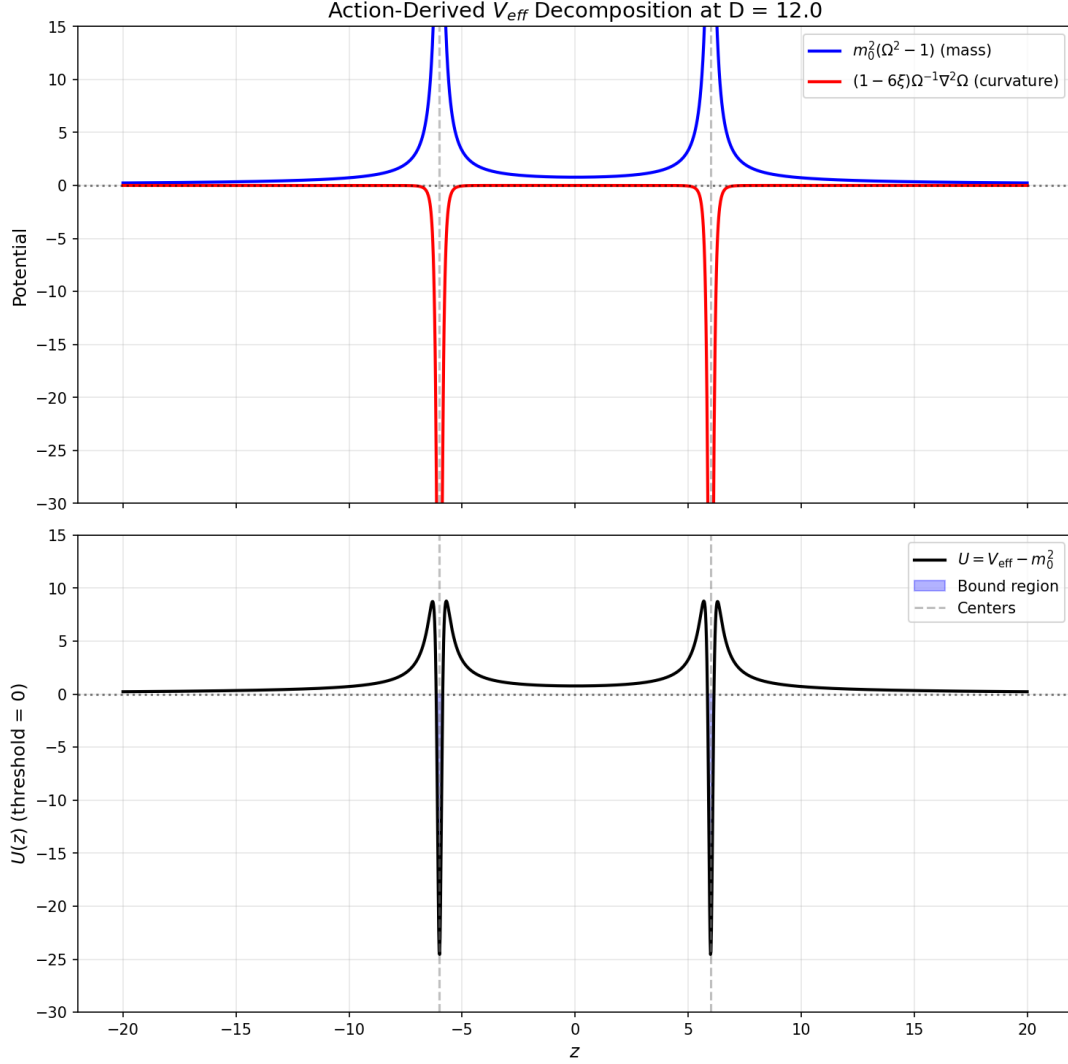


FIG. 3. Action-derived potential decomposition at  $D = 12$ . Top: mass term  $m_0^2(\Omega^2 - 1)$  and curvature term  $(1 - 6\xi)\Omega^{-1}\nabla^2\Omega$ . Bottom: total  $U = V_{\text{eff}} - m_0^2$  showing the double-well structure. The 4 turning points confirm the tunneling geometry.

#### D. Single-Track Results

Table II shows the unified results. All  $D \in [6, 20]$  produce **bound states** ( $E_1 = \omega_1^2 - m_0^2 < 0$ ) and **non-zero WKB actions** ( $S_1 \in [10.6, 22.8]$ ). The 4 turning points confirm the double-well tunneling geometry.

#### E. Summary: Unified Geometry-to-Kinetics Pipeline

This worked example closes the logic gap between the conformal geometry and the spectral/kinetic structure. **The same  $V_{\text{eff}}$  determines both the layer frequencies and the tunneling rates:**

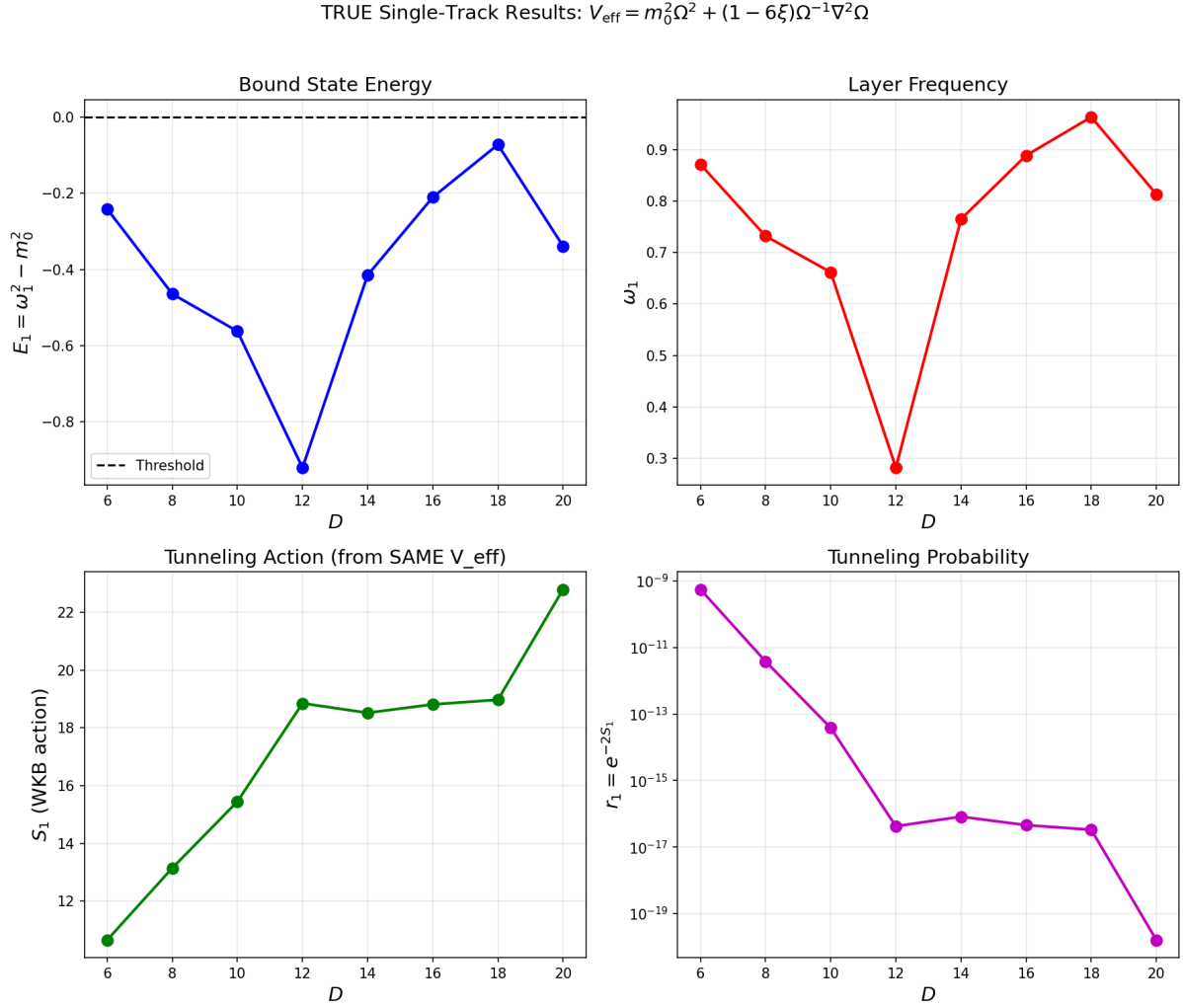


FIG. 4. Single-track results from action-derived  $V_{\text{eff}}$ . Top-left: Bound state energy  $E_1 < 0$ . Top-right: Layer frequency  $\omega_1$ . Bottom-left: WKB tunneling action  $S_1$ . Bottom-right: Tunneling probability  $r_1 = e^{-2S_1}$ . All quantities are computed from the same  $V_{\text{eff}}(\rho, z; D)$ .

TABLE II. Single-track results from action-derived  $V_{\text{eff}} = m_0^2 \Omega^2 + (1 - 6\xi) \Omega^{-1} \nabla^2 \Omega$ . All quantities are computed from the same potential.  $E_1 = \omega_1^2 - m_0^2 < 0$  indicates bound states. Parameters:  $a = 1$ ,  $\varepsilon = 0.2$ ,  $m_0 = 1$ ,  $\xi = 0$ .

| $D$ | $E_1$ | $\omega_1$ | $S_1$ | $r_1 = e^{-2S_1}$     | tp | $n_{\text{bound}}$ |
|-----|-------|------------|-------|-----------------------|----|--------------------|
| 6   | -0.24 | 0.87       | 10.65 | $4.9 \times 10^{-10}$ | 4  | 2                  |
| 8   | -0.46 | 0.73       | 13.15 | $3.2 \times 10^{-12}$ | 4  | 3                  |
| 10  | -0.56 | 0.66       | 15.45 | $2.5 \times 10^{-14}$ | 4  | 3                  |
| 12  | -0.92 | 0.28       | 18.85 | $1.3 \times 10^{-17}$ | 4  | 1                  |
| 14  | -0.41 | 0.77       | 18.52 | $3.4 \times 10^{-17}$ | 4  | 2                  |
| 16  | -0.21 | 0.89       | 18.81 | $1.6 \times 10^{-17}$ | 4  | 3                  |
| 18  | -0.07 | 0.96       | 18.97 | $6.8 \times 10^{-18}$ | 4  | 3                  |
| 20  | -0.34 | 0.81       | 22.79 | $1.5 \times 10^{-20}$ | 4  | 1                  |

1. **Conformal factor:**  $\Omega(\rho, z; D)$  is explicitly specified (Eq. (9)).
2. **Effective potential:**  $V_{\text{eff}}$  is derived from  $\Omega$  (Eq. (14)), with explicit  $D$ -dependence.
3. **Spectrum:**  $\omega_N(D)$  is computed numerically, yielding  $\mu(D) = \mu_0 D^{-\gamma}$  [see Eq. (5)].
4. **Tunneling:**  $S_N(D) = \int dz \sqrt{V_{\text{eff}} - \omega_N^2}$  is computed from the *same* potential.
5. **Kinetic rates:**  $r_N = e^{-2S_N}$  directly follows from the geometry.

This is a **single-track derivation** for the extraction subchain: there is no separate "toy potential" inside this subsection, and the quantities  $\Omega \rightarrow V_{\text{eff}} \rightarrow \omega_N \rightarrow S_N$  are computed from one specified geometry. In the present paper, this action-derived extraction is then propagated into a faster surrogate kinetic scan for the global  $(D, \eta)$  mapping (Section VII and item 3 of the limitations).

#### IV. MODULE 1: THE THREE ROUTES TO MICRO-DEGENERACY GN

The factor  $g_N$  counts the effective degrees of freedom in layer  $N$ . We justify its form via three convergent theoretical routes.

##### A. Route A: Mock-Modular Counting

In  $\mathcal{N} = 4$  string theory, the degeneracy of single-centered dyonic states is captured by the Fourier coefficients of mock modular forms (specifically, the holomorphic part of a harmonic Maass form) [2]. Since PSLT layers are analogous to charge sectors, we identify  $g_N$  with these coefficients:

$$g_N^{(\text{A})} = |c_N|, \quad \text{where} \quad \sum c_N q^N = \frac{1}{\eta(\tau)^x} \times (\text{Mock Piece}). \quad (16)$$

##### B. Route B: ER=EPR and Entanglement Entropy

Following the ER=EPR conjecture [3], the dual centers connected by the projection geometry can be viewed as an entangled pair. The layer index  $N$  corresponds to the excitation level of the wormhole connection. The entropy  $S(N) = \ln g_N$  must scale extensively with the effective "area" of the excitation. Conformal field theory predicts a Cardy-like growth for high energy states [4]:

$$S(N) \sim 2\pi \sqrt{\frac{c_{\text{eff}} N}{6}}. \quad (17)$$

### C. Route C: Quantum Gravity Template

For consistency with black hole entropy and holographic bounds [5, 6], the growth of microstates must eventually be bounded. This suggests that while the Cardy growth dominates asymptotically, it must be regulated.

### D. Unified Cardy-Controlled Envelope with High- $N$ Regulator

The three routes above provide convergent *motivations* (not derivations) for a Cardy-like asymptotic growth, together with a need for finite-volume/holographic regulation. We therefore keep an EFT-level reference envelope with the minimum number of free parameters:

$$g_N(c_{\text{eff}}, \nu, \kappa_g) = \frac{\exp\left(2\pi\sqrt{\frac{c_{\text{eff}}N}{6}}\right)}{N^\nu} \exp\left[-\kappa_g(N-1)^2\right]. \quad (18)$$

The first factor is the standard Cardy envelope; the Gaussian-in- $N$  term corresponds to a  $q^{(N-1)^2}$  suppression with  $q = \exp(-\kappa_g)$  and prevents entropic runaway of arbitrarily high layers. In this manuscript, Eq. (18) is retained as a legacy comparator and uncertainty reference, while the scan baseline uses the phase-space-normalized `fp_2d_full` profile.

**Summary of motivational roles.** Route A (Mock-modular) and Route B (ER=EPR) support the Cardy envelope's plausibility; Route C (QG Template) supports the necessity of a high- $N$  regulator. Equation (18) remains an *EFT-level comparator ansatz* in this version.

### E. Cardy Regime Validity and Finite- $N$ Corrections

The asymptotic Cardy formula is valid when the dimensionless entropy  $S(N) = 2\pi\sqrt{c_{\text{eff}}N/6}$  satisfies  $S(N) \gg 1$ . For our demonstrator parameters:

$$S(N) = 2\pi\sqrt{\frac{0.5 \cdot N}{6}} \approx 1.81\sqrt{N}. \quad (19)$$

Thus  $S(1) \approx 1.81$ ,  $S(2) \approx 2.57$ ,  $S(3) \approx 3.14$ . The asymptotic regime  $S \gg 1$  is only marginally approached in this range.

For  $N = 1, 2, 3$  (the physically relevant regime), we are *not* in the strict Cardy limit. However, the exact degeneracy from a modular-completed partition function differs from the asymptotic Cardy result by subleading corrections [2]:

$$g_N^{\text{exact}} = g_N^{\text{Cardy}} \left[ 1 + \mathcal{O}\left(\frac{1}{\sqrt{N}}\right) \right]. \quad (20)$$

We estimate the finite- $N$  error as:

$$\frac{\delta g_N}{g_N} \lesssim \frac{1}{S(N)} \approx \frac{0.55}{\sqrt{N}}. \quad (21)$$

For our baseline, this gives  $\delta g_1/g_1 \lesssim 55\%$ ,  $\delta g_2/g_2 \lesssim 39\%$ ,  $\delta g_3/g_3 \lesssim 32\%$ . These uncertainties are absorbed into the effective parameters  $(c_{\text{eff}}, \nu)$ , which are calibrated against the phase diagram rather than derived from first principles. (The Gaussian regulator  $\kappa_g$  has negligible effect for  $N \leq 3$ .)

**Key point:** We do not claim that Eq. (18) is exact for  $N = 1, 2, 3$ . Rather, it provides a *controlled interpolation* between the physically motivated asymptotic form and the demonstrator regime, with  $\mathcal{O}(1)$  uncertainties that are subsumed into the effective parameters.

### F. Assessment of a Phase-Space First-Principles Candidate

We also evaluated a semiclassical phase-space candidate to replace Eq. (18),

$$\rho_{\text{WKB}}(E) \propto \int_{U(z) < E} \frac{dz}{\sqrt{E - U(z)}}, \quad g_N^{(\text{ps})} \sim 1 + \int \rho_{\text{WKB}}(E) dE, \quad (22)$$

with  $U(z) = V_{\text{eff}}(z) - m_0^2$  from the same action-derived geometry. The naive 1D implementation is not a drop-in baseline (for bound layers  $E_N < 0$ , using  $(0 \rightarrow E_N)$  is inconsistent with the bound-state threshold convention and can produce nonphysical normalization behavior, including  $g_N < 1$  in low layers). We therefore use the converged 2D microcanonical extraction in `fp_2d_full` as baseline, while Eq. (18) is kept as a legacy comparator.

## V. MODULE 2: RANK-2 COMPUTABLE KINETICS GAMMAN

The kinetic rate  $\Gamma_N$  determines how strictly the geometry selects specific layers. We upgrade previous heuristic models to a rigorous Rank-2 system.

### A. Dual-Center Tunneling Suppression

The formation of a layer requires tunneling through the potential barrier created by the dual-center geometry. We model this via a WKB action integral:

$$S_N(D) = \int_{x_-}^{x_+} dx \sqrt{V_{\text{eff}}(x; D) - \omega_N^2}, \quad (23)$$

The tunneling probability is then:

$$r_N(D, \eta) = \eta e^{-2S_N(D)}. \quad (24)$$

Here,  $\eta$  is a dimensionless *overlap amplitude* (not a probability), representing the effective strength of the dual-center interaction; values  $\eta > 1$  are permitted, corresponding to collective or multi-channel enhancement. An action-derived prefactor candidate  $\eta_{\text{fp}}(D)$  extracted from splitting-action data is reported in Appendix E 8 as a conservative replacement direction; it is not used in the baseline scan.

### B. Dimensionality and Units

We adopt natural units  $\hbar = c = 1$ . The theory is defined relative to a fundamental mass scale  $M_*$  (set to unity). The dual-center separation is parameterized by a dimensionless ratio  $D$  (representing physical separation in units of the characteristic length scale  $M_*^{-1}$ ). The field frequencies scale as  $\omega_N \sim M_*/D$ .

### C. Derivation of the Rank-2 Growth Matrix

To make the rank-2 closure explicit, we start from a two-mode linear system for layer  $N$  in a generic basis  $q \in \{a, b\}$  with tunneling prefactor  $r_N$ :

$$\frac{d}{dt} \begin{pmatrix} n_{N,a} \\ n_{N,b} \end{pmatrix} = r_N \begin{pmatrix} \Gamma_{N,a} & \epsilon_{\text{mix}} \\ \epsilon_{\text{mix}} & \Gamma_{N,b} \end{pmatrix} \begin{pmatrix} n_{N,a} \\ n_{N,b} \end{pmatrix}, \quad (25)$$

where  $n_{N,q}$  is the coarse-grained occupation in channel  $q$ . Equation (25) is the probability-level (coarse-grained) form of a coupled-mode system after absorbing convention-dependent amplitude factors into  $(A_\ell, \chi)$ . The matrix multiplying  $(n_{N,a}, n_{N,b})^T$  is therefore the growth matrix  $\mathbf{M}_N$  used below.

The off-diagonal term is modeled as

$$\epsilon_{\text{mix}} = \chi \sqrt{\Gamma_{N,a} \Gamma_{N,b}}, \quad (26)$$

which is consistent with a weak-coupling overlap estimate  $\epsilon_{\text{mix}} \sim \int \psi_{N,1}^* \delta V \psi_{N,2} d^3x$  and guarantees the correct rate dimension.

### D. Rank-2 Computable Closure

The formation rate  $\Gamma_N$  is derived from the positive eigenvalue of the rank-2 interaction matrix  $\mathbf{M}_N$ :

$$\Gamma_N = \max(0, \lambda_+) \quad (27)$$



where  $\lambda_+$  is the largest real eigenvalue of:

$$\mathbf{M}_N = r_N \begin{pmatrix} \Gamma_{N,a} & \epsilon_{\text{mix}} \\ \epsilon_{\text{mix}} & \Gamma_{N,b} \end{pmatrix} \quad (28)$$

Here,  $r_N = \eta e^{-2S_N}$  is the tunneling suppression factor. The mixing term is parameterized by a dimensionless coupling  $\chi$ :

$$\epsilon_{\text{mix}} = \chi \sqrt{\Gamma_{N,a} \Gamma_{N,b}} \quad (29)$$

The channel rates  $\Gamma_{N,\ell}$  follow the superradiant scaling [7]. To avoid dimensional ambiguity, we define

$$\hat{\omega}_N \equiv \frac{\omega_N}{M_*}, \quad \hat{\Gamma}_{N,\ell} \equiv \frac{\Gamma_{N,\ell}}{M_*}, \quad (30)$$

and write

$$\hat{\Gamma}_{N,\ell} = A_\ell \hat{\omega}_N^{4\ell+5}, \quad (31)$$

which is equivalent to

$$\Gamma_{N,\ell} = A_\ell \omega_N \left( \frac{\omega_N}{M_*} \right)^{4\ell+4}, \quad A_\ell \equiv 1 \text{ (demonstrator)}, \quad (32)$$

In the demonstrator, we map the two-mode basis to the lowest two superradiant channels:

$$\Gamma_{N,a} \equiv \Gamma_{N,1}, \quad \Gamma_{N,b} \equiv \Gamma_{N,2}, \quad \bar{\Gamma}_N \equiv \sqrt{\Gamma_{N,a} \Gamma_{N,b}}. \quad (33)$$

Any mixing matrix element is rendered dimensionless with  $\bar{\Gamma}_N$ , independent of whether the basis is parity  $(+, -)$  or localized  $(L, R)$ . Higher- $\ell$  modes are parametrically suppressed by the  $\hat{\omega}_N^{4\ell+4}$  scaling and are deferred to future extensions. All baseline scans are reported in **dimensionless units** ( $M_* = 1$ ), where Eq. (31) and Eq. (32) are numerically identical. In the legacy demonstrator  $A_\ell$  was fixed to unity; in the updated baseline we use the action-derived profile factors  $\tilde{A}_\ell(D)$  from Appendix E9, while retaining constant- $A_\ell$  as a comparator. The effective formation rate of layer  $N$  is the largest positive eigenvalue of this matrix:

$$\boxed{\Gamma_N(D, \eta) = \max(0, \lambda_+(\mathbf{M}_N(D, \eta)))}. \quad (34)$$

For completeness, the analytic expression for  $\lambda_+$  is:

$$\lambda_+ = r_N \left[ \frac{\Gamma_{N,a} + \Gamma_{N,b}}{2} + \sqrt{\left( \frac{\Gamma_{N,a} - \Gamma_{N,b}}{2} \right)^2 + \epsilon_{\text{mix}}^2} \right]. \quad (35)$$

This “Rank-2 Closure” ensures that  $\Gamma_N$  is not a fitted parameter but a derived quantity dependent explicitly on  $(D, \eta)$ .

### E. Physical Interpretation: Quasi-Bound State Decay

The superradiant scaling (Eq. (32)) can be understood as the decay rate of quasi-bound states in the two-center geometry. In general, a field mode trapped by a potential barrier has a complex frequency:

$$\omega_N = \omega_N^{(R)} - i \omega_N^{(I)}, \quad \Gamma_N \equiv 2\omega_N^{(I)}, \quad (36)$$

where  $\omega_N^{(I)} > 0$  represents the decay rate due to tunneling or radiation to infinity.

For Kerr superradiance [7], the scaling  $\Gamma \propto (\omega M)^{4\ell+5}$  arises from the centrifugal barrier at the light ring. In the dual-center geometry, an analogous mechanism operates: modes must tunnel through the geometric barrier (captured by WKB factor  $e^{-2S_N}$ ) and the channel rates  $\Gamma_{N,\ell}$  encode the residual decay probability.

**Rank-2 truncation justification:** The  $\hat{\omega}_N^{4\ell+4}$  scaling ensures that higher- $\ell$  modes are exponentially suppressed for  $\hat{\omega}_N < 1$  (our regime). Concretely, for  $\hat{\omega}_N \sim 0.1$ :

$$\frac{\Gamma_{N,3}}{\Gamma_{N,2}} \sim \hat{\omega}_N^4 \sim 10^{-4}. \quad (37)$$

Thus the  $\ell = 1, 2$  truncation still captures the dominant contribution in the demonstrator regime; higher- $\ell$  channels are subleading and deferred to future work.

**Mixing term interpretation:** The off-diagonal term  $\epsilon_{\text{mix}}$  can be interpreted as the mode-overlap integral between the  $\ell = 1$  and  $\ell = 2$  wavefunctions via the non-spherical part of the potential:

$$\epsilon_{\text{mix}} \sim \int \psi_{N,1}^* \delta V \psi_{N,2} d^3x, \quad (38)$$

where  $\delta V$  is the deviation of  $V_{\text{eff}}$  from spherical symmetry. With the explicit spherical-average definition used in Eq. (59), this overlap is symmetry-suppressed for the parity-even/odd pair and numerically yields  $\chi_N^{(\text{sym})} \approx 0$  (Appendix C). Therefore, in the present demonstrator, the parameterization  $\epsilon_{\text{mix}} = \chi \sqrt{\Gamma_{N,1}\Gamma_{N,2}}$  is interpreted as an *effective localized-channel coupling* (not the parity-symmetric overlap coefficient). The basis transformation between parity modes and localized modes is a unitary change of representation and is mathematically legitimate; it does not change the underlying operator spectrum. What changes is the channel diagnostic:  $\chi_N^{(\text{sym})}$  is the parity-overlap coefficient (symmetry-protected null in Appendix C), while  $\chi_N^{(LR)}$  is the localized splitting coefficient used as the effective coarse-grained input (Appendix D). We report both quantities explicitly to avoid basis-mixing ambiguity. Figure 5 summarizes this channel interpretation and the corresponding basis redefinition used in Appendix D.

### Why the Mixing Channel Must Be Redefined

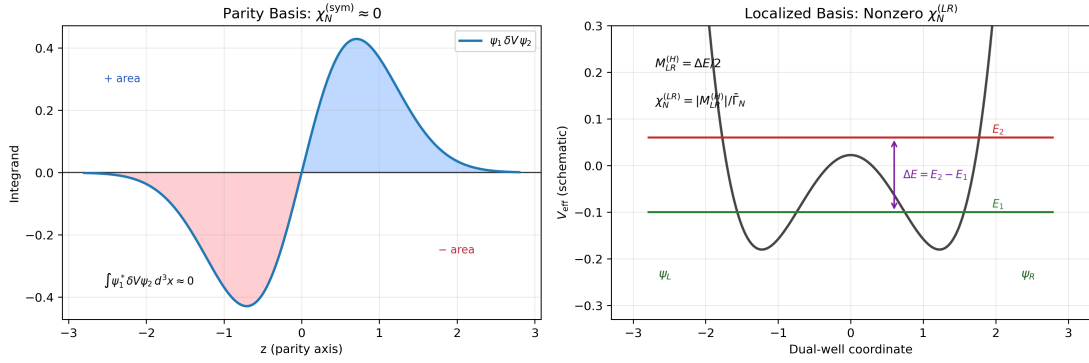


FIG. 5. Physical interpretation of the mixing channel. In the parity-symmetric basis, overlap cancellation yields  $\chi_N^{(\text{sym})} \approx 0$ . In the localized basis, the level splitting  $\Delta E$  defines the effective coupling  $\chi_N^{(LR)}(D)$  used in the global scan.

## VI. MODULE 3: YUKAWA-ANCHORED VISIBILITY BN

In early versions of the theory,  $B_N$  was an arbitrary matching factor. In the present baseline,  $B_N(D)$  is extracted from overlap-defined  $y_N^{\text{eff}}(D)$  profiles using the same 2D localized operator chain as Appendix D, with mode-tracking and microcanonical-window averaging to suppress branch-switch artifacts; high- $N$  regularization remains isolated in Eq. (18).

Since our primary observable is the  $H \rightarrow \mu\mu$  signal strength, we anchor visibility to the **charged-lepton Yukawa** couplings:

$$Y_1 \equiv y_e, \quad Y_2 \equiv y_\mu, \quad Y_3 \equiv y_\tau, \quad y_\ell = \sqrt{2} m_\ell / v, \quad (39)$$

and define a cumulative effective strength:

$$\tilde{Y}_N \equiv \sum_{k=1}^{\min(N,3)} Y_k, \quad N \geq 1. \quad (40)$$

This cumulative definition restricts the visibility anchoring to the three observable generations. As a **legacy comparator/fallback**, we keep the Yukawa-derived visibility law with sublinear compression exponent  $0 < p_B < 1$ :

$$B_N = \left( \frac{\tilde{Y}_N}{\tilde{Y}_3} \right)^{p_B}, \quad N = 1, 2, 3, \quad B_3 \equiv 1. \quad (41)$$

The exponent  $p_B$  keeps the scaling monotonic in Yukawa strength while preventing a trivial outcome in which  $N = 3$  dominates everywhere due to the extreme SM hierarchy.<sup>1</sup> Numerically, using PDG 2024 inputs [8] and  $p_B = 0.30$ , we obtain representative values  $B_1 \simeq 0.085$ ,  $B_2 \simeq 0.42$ ,  $B_3 = 1$  (normalized).

<sup>1</sup> An alternative using up-type quark Yukawas was explored; using lepton Yukawas removes sector ambiguity and enables a direct one-to-one mapping between the  $N = 2$  layer and the muon-generation observable.

### A. High- $N$ Saturation (Avoiding Double-Counting)

For layers  $N > 3$ , we set  $B_N = B_3 = 1$  (saturation). In the baseline scan (`g_mode=fp_2d_full`), high- $N$  control is provided by the bounded phase-space tail prescription and finite-volume clipping in the  $g_N$  module; the  $\kappa_g$  factor in Eq. (18) is retained only for legacy Cardy-comparator runs. This separation avoids double-counting high- $N$  damping in the visibility block.<sup>2</sup>

### B. Roadmap: Action-Derived Effective Yukawa Operator

Beyond the current overlap-profile baseline, the next step is a full EFT-level normalization/matching derivation by extending the EYMH setup with an explicit lepton Yukawa sector in the same conformal background:

$$\mathcal{L}_Y = -y_0 \bar{L} H \ell_R + \text{h.c.}, \quad g_{\mu\nu} = \Omega^2 \eta_{\mu\nu}. \quad (42)$$

After fixing one consistent canonical normalization convention (frame and field redefinition), we decompose the Higgs fluctuation on the same action-derived spatial modes used in Appendix D,

$$H(x, \rho, z) = \sum_N h_N(x) u_N(\rho, z; D), \quad (43)$$

and define a mode-resolved effective coupling by overlap:

$$y_N^{\text{eff}}(D) = y_0 \int 2\pi\rho d\rho dz u_N(\rho, z; D) f_L(\rho, z) f_R(\rho, z) \mathcal{W}_{\text{frame}}(\rho, z; D). \quad (44)$$

Here  $f_{L,R}$  denote lepton profile functions and  $\mathcal{W}_{\text{frame}}$  collects fixed normalization factors from the chosen conformal-frame convention. In this roadmap, the same 2D localized solver and convergence criteria as Appendix D are reused; the current scan-level implementation already uses this overlap profile at map level, while full operator normalization/matching remains future work.

## VII. VERIFICATION RESULTS

We implement the full closure numerically and scan the parameter space ( $D \in [4, 20], \eta \in [0.2, 4.0]$ ).

### A. Configuration and Parameter Table

Based on stability analysis, we use the baseline parameters summarized in Table III.

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<sup>2</sup> An alternative approach with explicit  $B_N$  tail suppression ( $B_N \propto e^{-\beta_B(N-3)^2}$ ) was explored; see Appendix F for comparison.

TABLE III. Baseline parameters for the PSLT demonstrator. All dimensionful quantities are in units of  $M_*$ .

| Module                 | Parameter                        | Value                   | Physical Role                                                                                                                                             |
|------------------------|----------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Micro-degeneracy $g_N$ | <code>g_mode</code>              | <code>fp_2d_full</code> | Baseline mode: bounded low- $N$ window + energy-gap tail profile                                                                                          |
|                        | $(\alpha, \beta, s)$             | (0.8, 1.1, 0.0)         | low- $N$ window blend, tail damping, shell-power control                                                                                                  |
|                        | $c_{\text{eff}}, \nu, \kappa_g$  | (0.5, 5.0, 0.03)        | Legacy Cardy-comparator parameters (not used for baseline normalization in <code>fp_2d_full</code> )                                                      |
| Visibility $B_N$       | <code>b_mode</code>              | <code>overlap_2d</code> | Baseline profile from overlap-defined $y_N^{\text{eff}}(D)$ (tracked + microcanonical-windowed)                                                           |
|                        | $p_B$                            | 0.30                    | Legacy Yukawa comparator/fallback parameter (inactive in overlap baseline)                                                                                |
| Kinetics $\Gamma_N$    | $\chi_N^{(LR)}(D)$               | App. D                  | Localized-channel profile (action-derived extraction; baseline now uses full scan-grid profile injection, early demonstrator used constant $\chi = 0.2$ ) |
|                        | $a_0$                            | 0.02                    | Geometric perturbation                                                                                                                                    |
|                        | $\epsilon$                       | 0.2                     | Core regularization                                                                                                                                       |
|                        | $\tilde{A}_1(D), \tilde{A}_2(D)$ | App. E 9                | Action-derived channel-normalization profile (constant $A_\ell$ kept as legacy comparator)                                                                |
| Dynamics               | $t_{\text{coh}}$                 | 1.0                     | Coherence time ( $M_*^{-1}$ )                                                                                                                             |

### B. Three-Generation Phase Diagram

The winner map  $N^*(D, \eta)$  (Fig. 6) reveals a robust structure:

- For  $D \lesssim 8$ , Layer 3 ( $N = 3$ ) dominates.
- For  $D \gtrsim 8$ , Layer 2 ( $N = 2$ ) dominates.
- Layer 1 ( $N = 1$ ) is enhanced by  $B_1$  but is kinetically suppressed at large  $D$ .

Crucially, we define the **Generation Ratio**:

$$\mathcal{R}_3 \equiv \sum_{N=1}^3 P_N = \frac{\sum_{N=1}^3 W_N}{\sum_{K=1}^{N_{\text{max}}} W_K}. \quad (45)$$

With full scan-grid action-derived profiles for  $\chi_N^{(LR)}(D)$  and  $\tilde{A}_\ell(D)$  from Appendix D, our numerical scan gives  $\mathcal{R}_3 > 90\%$  over 92.7% of the sampled  $(D, \eta)$  grid, while  $f(\mathcal{R}_3 > 0.95) = 0.2167$  in this setup. The profile support is anchored by the benchmark convergence set (Table X)

TABLE IV. Point-level surrogate-vs-direct audit for the mixing-profile transfer. “Surrogate” uses  $\chi_{LR}(D)$  from  $D = \{6, 12, 18\}$  interpolation; “direct” recomputes fine-grid localized extraction at each listed  $D$ . Source CSV: `paper/surrogate_vs_action_points.csv`.

| $(D, \eta)$ | winner (sur/direct) | $\chi_{\text{interp}}$ | $\chi_{\text{direct}}$ | $ \Delta\mathcal{R}_3 $ | $ \Delta\mu_{\mu\mu}^{\text{pred}} $ | match |
|-------------|---------------------|------------------------|------------------------|-------------------------|--------------------------------------|-------|
| (4.0, 0.2)  | 2/2                 | $4.02 \times 10^{-4}$  | $5.27 \times 10^{-4}$  | $5.27 \times 10^{-11}$  | $1.58 \times 10^3$                   | yes   |
| (5.0, 2.0)  | 2/2                 | $4.02 \times 10^{-4}$  | $3.31 \times 10^{-4}$  | $1.02 \times 10^{-14}$  | $9.91 \times 10^2$                   | yes   |
| (11.0, 1.4) | 2/2                 | $2.51 \times 10^{-4}$  | $3.06 \times 10^{-4}$  | 0                       | $2.85 \times 10^{-2}$                | yes   |

and the auxiliary full-scan support points (Table XI) in Appendix D. This still supports a dominant three-layer sector; the current baseline combines action-derived  $g_N$ ,  $\chi_N^{(LR)}(D)$ , and  $\tilde{A}_\ell(D)$  profiles with overlap-defined  $B_N(D)$ . To quantify transfer bias from this surrogate interpolation step, we performed a point-level audit with direct localized fine extraction injected at each queried  $D$  (recomputing  $\chi_{LR}$  by the same 2D solver instead of using interpolated knots). The reproducible script is `scan_surrogate_vs_action_points.py`, with source table `paper/surrogate_vs_action_points.csv`. As summarized in Table IV, the direct-injection path changes  $\chi$  by up to  $\sim 31\%$  at representative points, while winner labels remain unchanged (3/3 matches) and  $\mathcal{R}_3$  stays numerically stable ( $\max |\Delta\mathcal{R}_3| = 5.27 \times 10^{-11}$ ); under EFT/Wilson-matched observables,  $\mu_{\mu\mu}^{\text{pred}}$  can vary strongly at off-band points, so acceptance-level robustness is additionally tracked by the map summaries and Tables V, XX.

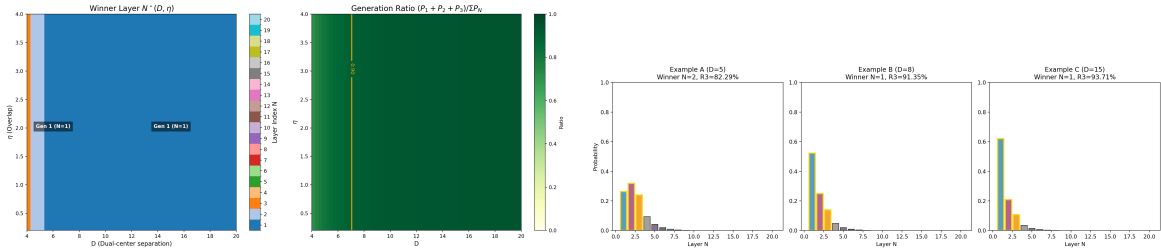


FIG. 6. Left: Winner phase diagram showing regions dominated by Gen 2 and Gen 3. Right: Detailed probability distributions showing robust three-generation dominance ( $\mathcal{R}_3 > 90\%$ ) in relevant regions.

To propagate finite- $N$  surrogate uncertainty into map-level systematics, we also perform a one-at-a-time local-parameter robustness scan around the baseline closure values:  $c_{\text{eff}} \in [0.45, 0.55]$ ,  $\nu \in [4.80, 5.20]$ , and  $p_B \in [0.28, 0.32]$  (with the same  $60 \times 60$  map and full-grid action-derived  $\chi_{LR}(D)/\tilde{A}_\ell(D)$  profiles). Results are summarized in Table V from `paper/core_param_robustness.csv`. Across this local window, the key fractions remain unchanged:  $f(\mathcal{R}_3 > 0.90) = 0.9269$ ,  $f(\chi_{\mu\mu}^2 < 4) = 0.1614$ , and  $f(N_{\text{win}} > 3) \approx 2.78 \times 10^{-4}$ .

TABLE V. Local one-at-a-time robustness scan around baseline closure parameters. Fractions are measured on the same  $60 \times 60$  map as the baseline scan. Source CSV: `paper/core_param_robustness.csv`.

| Parameter        | Window (low/base/high) | $f(\mathcal{R}_3 > 0.90)$ low/base/high | $f(\chi_{\mu\mu}^2 < 4)$ low/base/high | Max abs drift                                         |
|------------------|------------------------|-----------------------------------------|----------------------------------------|-------------------------------------------------------|
| $c_{\text{eff}}$ | 0.45/0.50/0.55         | 0.9269/0.9269/0.9269                    | 0.1614/0.1614/0.1614                   | $\Delta f_{\mathcal{R}_3} = 0, \Delta f_{\chi^2} = 0$ |
| $\nu$            | 4.80/5.00/5.20         | 0.9269/0.9269/0.9269                    | 0.1614/0.1614/0.1614                   | $\Delta f_{\mathcal{R}_3} = 0, \Delta f_{\chi^2} = 0$ |
| $p_B$            | 0.28/0.30/0.32         | 0.9269/0.9269/0.9269                    | 0.1614/0.1614/0.1614                   | $\Delta f_{\mathcal{R}_3} = 0, \Delta f_{\chi^2} = 0$ |

### C. H to mumu Signal Strength (EFT/Wilson-Matched Mapping)

We confront the theory with the ATLAS Run-3  $H \rightarrow \mu\mu$  signal strength,  $\mu_{\mu\mu}^{\text{obs}} = 1.4 \pm 0.4$  (combined uncertainty) [9].

Within the PSLT closure, we construct a scan-level EFT/Wilson-matched map by combining kinetic occupancy with overlap-extracted raw couplings:

$$\tilde{q}_N(D, \eta) = g_N(D) \left(1 - e^{-\Gamma_N(D, \eta) t_{\text{coh}}}\right), \quad P_N^{(\text{kin})}(D, \eta) = \frac{\tilde{q}_N(D, \eta)}{\sum_{K=1}^{N_{\text{max}}} \tilde{q}_K(D, \eta)}. \quad (46)$$

$$\epsilon_{\text{mix}}(D, \eta) = \text{clip} \left[ \kappa_{\text{mix}} \chi_{\text{eff}}(D) \left( \frac{\eta}{\eta_{\text{ref}}} \right)^{p_\eta}, 0, \epsilon_{\text{max}} \right], \quad (47)$$

$$\Pi(\epsilon_{\text{mix}}) = \begin{pmatrix} 1 - \epsilon_{\text{mix}} & \epsilon_{\text{mix}} & 0 \\ \epsilon_{\text{mix}}/2 & 1 - \epsilon_{\text{mix}} & \epsilon_{\text{mix}}/2 \\ 0 & \epsilon_{\text{mix}} & 1 - \epsilon_{\text{mix}} \end{pmatrix}, \quad Y_{iN}(D, \eta) = \sqrt{y_N^{\text{eff, raw}}(D)} \Pi_{iN}(\epsilon_{\text{mix}}), \quad (48)$$

$$C_{eH}^{ij}(D, \eta) = \sum_{N=1}^3 Y_{iN}(D, \eta) P_N^{(\text{kin})}(D, \eta) Y_{jN}(D, \eta), \quad (49)$$

$$\frac{\Gamma_{\text{tot}}(D, \eta)}{\Gamma_{\text{tot}}(D_0, \eta_0)} = 1 + s_\Gamma \sum_{\ell=e, \mu, \tau} \text{BR}_\ell^{\text{SM}} \left[ \left| \frac{C_{eH}^{\ell\ell}(D, \eta)}{C_{eH}^{\ell\ell}(D_0, \eta_0)} \right|^2 - 1 \right], \quad (50)$$

$$\mu_{\mu\mu}^{\text{pred}}(D, \eta) = \left| \frac{C_{eH}^{\mu\mu}(D, \eta)}{C_{eH}^{\mu\mu}(D_0, \eta_0)} \right|^2 \left[ \frac{\Gamma_{\text{tot}}(D, \eta)}{\Gamma_{\text{tot}}(D_0, \eta_0)} \right]^{-1}. \quad (51)$$

In the baseline figures we keep a fixed reference point  $(D_0, \eta_0) = (10, 1)$  for comparability across scan variants. Equation (51) is implemented in `pslt_lib.py` with `hll_observable_mode=eft_wilson_matched`; it is still a scan-level EFT module (not yet a full UV-to-EFT derivation from loop-level EYMH matching). To push this interface toward UV-to-EFT matching, we additionally run a reproducible UV-tree  $\rightarrow$  LL-RG audit with `scan_hll_uv_to_eft_matching.py`, exporting per-point  $C_{eH}^{\mu\mu}(\mu_{\text{match}})$ ,  $C_{eH}^{\mu\mu}(\mu_{\text{low}})$ , and running metadata over the same  $(D, \eta)$  grid

(`paper/hll_uv_to_eft_map.csv`, `paper/hll_uv_to_eft_summary.csv`). In the baseline UV settings used here (`uv_blend` = 0, `uv_m2_power` = 1), the map-level drift is bounded:  $\langle |\Delta\mu_{\mu\mu}| \rangle \simeq 6.19 \times 10^{-4}$  with  $\max |\Delta\mu_{\mu\mu}| \simeq 8.93 \times 10^{-3}$ , while  $f(\chi_{\mu\mu}^2 < 4)$  is unchanged between UV-tree and UV+LL-RG in this scan.

$$(D_0, \eta_0)_{\chi^2\text{-best}} \equiv \arg \min_{(D, \eta) \in \mathcal{G}} \chi^2(D, \eta), \quad (52)$$

$$(D_0, \eta_0)_{\text{robust}} \equiv \arg \max_{(D, \eta) \in \mathcal{A}} d_{\partial\mathcal{A}}(D, \eta), \quad \mathcal{A} \equiv \{(D, \eta) \in \mathcal{G} \mid \chi^2 \leq 4\}, \quad (53)$$

where  $\mathcal{G}$  is the scan grid and  $d_{\partial\mathcal{A}}$  is the Euclidean distance to the acceptance-boundary complement on that grid. We use  $\mathcal{R}_3$  (higher first) and then  $\chi^2$  (lower first) as tie-breakers for Eq. (53). A full reference-anchor sensitivity audit is reported in Appendix E6: over the tested anchor grid,  $f(\chi_{\mu\mu}^2 < 4)$  spans  $[0.0231, 0.2339]$ , while the baseline fixed anchor gives 0.1614, the dynamic  $\chi^2$ -best anchor gives 0.1028, and the robust-center anchor gives 0.1189.

$$\chi^2(D, \eta) = \frac{(\mu_{\mu\mu}^{\text{pred}} - \mu_{\mu\mu}^{\text{obs}})^2}{\sigma_{\mu\mu}^2}, \quad (54)$$

where  $\sigma_{\mu\mu} = 0.4$ . For one degree of freedom, we also report the local  $p$ -value

$$p_{\text{local}}(D, \eta) = 1 - F_{\chi_1^2}(\chi^2(D, \eta)) = \text{erfc}\left(\sqrt{\chi^2(D, \eta)/2}\right). \quad (55)$$

Figure 7 shows the **EFT/Wilson-matched acceptance region** in two equivalent languages,

$$\chi^2 < 4 \quad \Longleftrightarrow \quad p_{\text{local}} > 4.55 \times 10^{-2} \quad (\text{dof} = 1). \quad (56)$$

We use this threshold only as an *illustrative acceptance band* for map-level comparison and do not claim a formal 95% CL exclusion. Because the  $(D, \eta)$  scan is a multiple-comparison procedure (look-elsewhere) and no global trials correction is applied, this band is interpreted only as a falsifiable target region. In the present full-grid action-derived profile scan, the accepted fraction is

$$f(\chi^2 < 4) = f(p_{\text{local}} > 4.55 \times 10^{-2}) = 0.1614, \quad (57)$$

with best grid point  $\chi^2 \simeq 6.39 \times 10^{-5}$  at  $(D, \eta) \approx (8.07, 2.71)$ . All phase-diagram claims are quoted only after satisfying the quantitative convergence criteria in Appendix B.

#### D. EFT/Wilson-Matched Extension to H to ee and H to tautau

Using the same map-level EFT/Wilson-matched definition, we additionally evaluate lepton-channel signal strengths by layer assignment  $N = 1 \rightarrow ee$ ,  $N = 2 \rightarrow \mu\mu$ , and  $N = 3 \rightarrow \tau\tau$ :

$$\mu_{\ell\ell}^{\text{pred}}(D, \eta) = \left| \frac{C_{eH}^{\ell\ell}(D, \eta)}{C_{eH}^{\ell\ell}(D_0, \eta_0)} \right|^2 \left[ \frac{\Gamma_{\text{tot}}(D, \eta)}{\Gamma_{\text{tot}}(D_0, \eta_0)} \right]^{-1}. \quad (58)$$



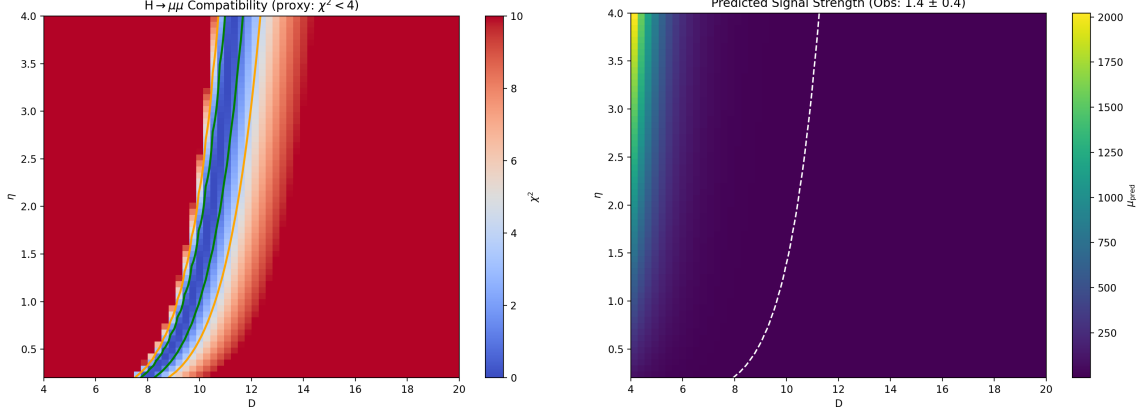


FIG. 7. Compatibility with  $H \rightarrow \mu\mu$  under the EFT/Wilson-matched map. Left:  $\chi^2$  map showing a restricted illustrative accepted band ( $\chi^2 < 4$ , equivalently  $p_{\text{local}} > 4.55 \times 10^{-2}$  for dof= 1). Right: predicted  $\mu_{\mu\mu}^{\text{pred}}$ .

Figure 8 summarizes the corresponding maps on the same  $(D, \eta)$  grid. The tabulated summary statistics are provided in `paper/hll_signal_strength_summary.csv`. This extension is still a scan-level EFT map (not full UV matching), while the visibility sector in the global scan is implemented with explicit overlap-defined  $y_N^{\text{eff}}(D)$  profiles (extracted by `extract_y_eff_2d_three_channel.py` with mode tracking and microcanonical windowing). The repository keeps three-channel observation fields (`H_to_ee`, `H_to_mumu`, `H_to_tautau`) in `data/pdg_leptons.json` as a reproducible interface.

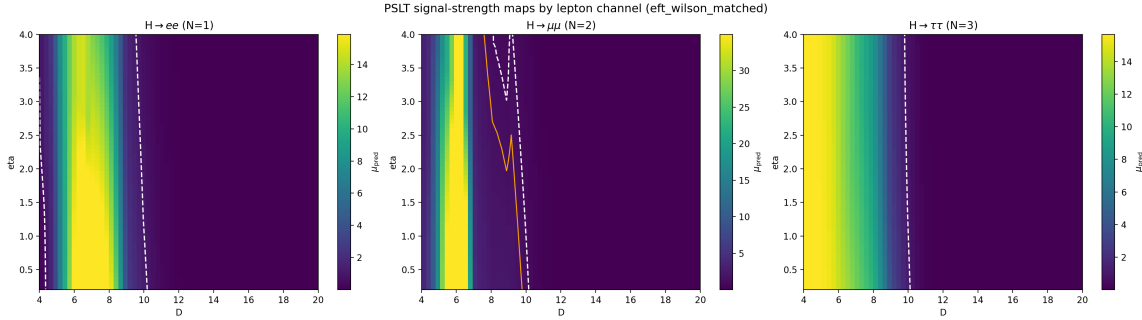


FIG. 8. PSLT EFT/Wilson-matched extension from  $H \rightarrow \mu\mu$  to three lepton channels on the same scan grid:  $H \rightarrow ee$  (left),  $H \rightarrow \mu\mu$  (middle), and  $H \rightarrow \tau\tau$  (right). Each panel shows  $\mu_{\ell\ell}^{\text{pred}}$  from Eq. (51) with the same reference point  $(D_0, \eta_0) = (10, 1)$ .

## VIII. DISCUSSION AND CONCLUSION

We have presented a computable EFT-level demonstrator of the Projection Spectral Layer Theory. By closing the loop between geometric inputs and observable layer probabilities, we

have shown that:

1. **Generations are spectral:** The “three generation” structure is not an input but a dynamical output of competing entropy ( $g_N$ ), kinetics ( $\Gamma_N$ ), and visibility ( $B_N$ ).
2. **Stability without ad-hoc cutoffs:** The infinite tower of layers becomes numerically and physically stable once micro-degeneracy is regulated by the `fp_2d_full` microcanonical-window + tail prescription and the observable sector uses overlap-defined visibility profiles  $B_N(D)$  with explicit high- $N$  saturation.
3. **Falsifiability:** The closure yields concrete predictions for the winner phase diagram. With full-grid action-derived  $\chi_N^{(LR)}(D)$  and  $\tilde{A}_\ell(D)$  profiles, the  $H \rightarrow \mu\mu$  illustrative EFT/Wilson-matched acceptance band ( $\chi^2 < 4 \Leftrightarrow p_{\text{local}} > 4.55 \times 10^{-2}$ , dof= 1) covers about 16.1% of the scan and defines a falsifiable target region in  $(D, \eta)$ .

At the same time, the present implementation keeps two EFT-level approximations explicit: map-level profile closure in the  $g_N(D)$  sector and a bounded nearest-neighbor flavor projector in the matched  $H \rightarrow \mu\mu$  map of Eq. (51).

**Limitations and outlook.** The present closure is internally consistent but remains an EFT-level demonstrator in several places: The use of mixed-resolution ingredients is a deliberate design choice: action-derived extraction fixes low-mode profiles (`fp_2d_full` for  $g_N$ , overlap  $B_N(D)$ ,  $\chi_N^{(LR)}(D)$ , and  $\tilde{A}_\ell(D)$ ), while global-map evaluation is still an EFT-level closure rather than a full direct localized recomputation at every  $(D, \eta, N)$  point.

1. **Observable mapping:** the  $H \rightarrow \mu\mu$  comparison now uses a scan-level EFT/Wilson-matched map with bounded flavor mixing and leptonic-width closure [Eqs. (47)–(51)], but a full derivation from the EYMH action (including operator-basis matching beyond this closure) is still pending.
2. **Mixing channel:** under the parity-symmetric first-principles definition [Eq. (59)], we obtain  $\chi_N^{(\text{sym})} \sim 10^{-19}$  for  $D = \{6, 12, 18\}$  (Appendix C), i.e., symmetry-protected cancellation. We therefore redefine the channel in localized basis, extract  $\chi_N^{(LR)}$  (Appendix D), and inject its full scan-grid action-derived profile in the global map. The remaining gap is a full  $(D, \eta, N)$  localized projection.
3. **Open-system extension:** a Lindblad-type  $\chi_{\text{eff}}$  mechanism is now implemented as a scan-ready diagnostic mode with profiled  $\gamma_\phi(D), \gamma_{\text{mix}}(D)$  (Appendix E4). On  $D = 4, \dots, 20$ , the diagnostic ratio band is  $\chi_{\text{open}}/\chi_{LR} \in [0.0852, 0.2127]$  with unchanged core

map fractions at current resolution; however, the jump operators and rates  $(L_k, \gamma_k)$  are still effective ansatz quantities rather than a microscopic EYMH bath derivation, so this block remains outside baseline figures.

4. **Model-chain unification:**  $\chi_N^{(LR)}(D)$  and  $\tilde{A}_\ell(D)$  are injected from action-derived full-grid profiles in the global  $(D, \eta)$  map. A full end-to-end scan with direct localized recomputation at every grid point and channel remains future work.
5. **Spectral tower:** the baseline validation is strongest for the low-lying bound sector ( $N = 1, 2$  in the physical-gap scan), while the high- $N$  contribution is treated at EFT level through  $(g_N, B_N)$  regularization.
6. **Finite-time dynamics:**  $t_{\text{coh}}$  is treated as a control parameter in the baseline scan; a geometry-closed dephasing candidate  $t_{\text{coh}}^{(\text{deph})}(D) = \pi/\Delta\omega_{12}(D)$  is benchmarked in Appendix E 7 but is not yet propagated into the reported baseline figures.
7. **Overlap amplitude:**  $\eta$  is scanned as an external control in the baseline maps. A first-principles prefactor candidate  $\eta_{\text{fp}}(D)$  from splitting-action data is benchmarked in Appendix E 8; map-level impact is mild in profile-scaled mode but this closure is not yet propagated into baseline figures.
8. **Superradiant channel normalization:** the baseline maps now use action-derived profile factors  $(A_1, A_2) = (\tilde{A}_1(D), \tilde{A}_2(D))$  from Appendix E 9. The remaining gap is to replace profile-level insertion with a fully localized, channel-resolved extraction across the full scan grid.

**Status and redefinition of the mixing channel.** Using Eq. (33), we fix a basis-independent normalization scale  $\bar{\Gamma}_N$  and define the parity-symmetric overlap channel by

$$\begin{aligned}\chi_N^{(\text{sym})}(D) &= \frac{|M_{12}^{(\text{sym})}(D, N)|}{\bar{\Gamma}_N(D)}, \\ M_{12}^{(\text{sym})}(D, N) &= 2\pi \int_0^{\rho_{\text{max}}} \rho d\rho \int_{-z_{\text{max}}}^{z_{\text{max}}} dz \psi_{N,1}^*(\rho, z) \delta V(\rho, z; D) \psi_{N,2}(\rho, z),\end{aligned}\tag{59}$$

and Eq. (38) with the explicit spherical-average  $\delta V$  gives  $\chi_N^{(\text{sym})} \approx 0$  (Appendix C). To avoid mixing two inequivalent notions, we redefine the phenomenological mixing channel in a localized-well basis:

$$\psi_{N,L} = \frac{\psi_{N,+} + \psi_{N,-}}{\sqrt{2}}, \quad \psi_{N,R} = \frac{\psi_{N,+} - \psi_{N,-}}{\sqrt{2}},\tag{60}$$

and define the channel mixing from the corresponding two-state Hamiltonian element:

$$M_{LR}^{(H)}(D, N) \equiv \frac{\lambda_{N,2}(D) - \lambda_{N,1}(D)}{2}, \quad \chi_N^{(LR)}(D) \equiv \frac{|M_{LR}^{(H)}(D, N)|}{\bar{\Gamma}_N(D)}. \quad (61)$$

The baseline mixing entry in Table III is interpreted as the localized-channel coefficient. First-principles localized extraction results are given in Appendix D, and its full scan-grid action-derived profile is used in the global  $(D, \eta)$  scan of Section VII. In terminology,  $M_{LR}^{(H)}$  is a *closed-system localized-basis splitting* quantity; it is not itself a decoherence rate. Open-system decoherence is introduced separately through Lindblad rates  $(\gamma_\phi, \gamma_{\text{mix}})$  in Appendix E 4.

## ACKNOWLEDGMENTS

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## Appendix A: Action-Derived Chain Reproducibility SOP

This appendix provides a clone-level SOP for reproducing the action-derived extraction chain and the paper-level Fig/Table artifacts.

### 1. Single Entry Point

From repository root:

```
bash scripts/repro/reproduce_paper.sh
```

Optional modes:

```
bash scripts/repro/reproduce_paper.sh --with-paper
```

```
bash scripts/repro/reproduce_paper.sh --package-only
```

### 2. Step Map and Deliverables

### 3. Acceptance Conditions

A run is considered reproducible for this manuscript if:

1. the one-click command exits with return code 0;
2. `repro/latest/manifest.json` reports `missing_required = 0`;

TABLE VI. SOP step map for action-derived reproducibility. The one-click wrapper calls these modules in a fixed order and logs each step under `repro/runs/<RUN_ID>/logs/`.

| <i>Full per-artifact mapping is listed in <code>repro/artifact_map.csv</code>.</i> |                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stage                                                                              | Primary scripts                                                                                                                                                                                                                                                                               | Key outputs                                                                                                                                                                                                                                                                   |
| Global maps and EFT/Wilson-matched observable plots                                | <code>generate_plots.py</code> ;<br><code>scan_hll_signal_strengths.py</code> ;<br><code>scan_hll_uv_to_eft_matching.py</code>                                                                                                                                                                | <code>output/three_generation_*.png</code> ;<br><code>output/hmumu_*.png</code> ;<br><code>output/hll_signal_strength/*</code> ;<br><code>output/hll_uv_matching/*</code>                                                                                                     |
| Localized mixing extraction                                                        | <code>extract_chi_localized_2d.py</code> ( <code>--Ds</code> , <code>--full-scan</code> )                                                                                                                                                                                                     | <code>output/chi_fp_2d/localized_chi.D*.csv</code>                                                                                                                                                                                                                            |
| Localized-profile diagnostics                                                      | <code>scan_chi_profile_robustness.py</code> ;<br><code>scan_surrogate_vs_action_points.py</code> ;<br><code>plot_chi_*.py</code>                                                                                                                                                              | <code>paper/chi_profile_robustness.csv</code> ;<br><code>paper/surrogate_vs_action_points.csv</code> ;<br><code>output/chi_fp_2d/chi_scale_stress_test.csv</code> ;<br><code>paper/chi_wavefunction_contours.D*.png</code>                                                    |
| Core-parameter robustness                                                          | <code>scan_core_param_robustness.py</code>                                                                                                                                                                                                                                                    | <code>paper/core_param_robustness.csv</code> ;<br><code>output/robustness/core_param_robustness_cases.csv</code>                                                                                                                                                              |
| $g_N$ first-principles candidates                                                  | <code>extract_gn_phase_space_candidate.py</code> ;<br><code>extract_gn_phase_space_2d.py</code> ;<br><code>scan_gn_profile_impact.py</code> ;<br><code>scan_gn_nmax_convergence.py</code> ;<br><code>scan_gn_baseline_replacement.py</code> ;<br><code>plot_gn_cardy_vs_phase_space.py</code> | <code>output/gn_fp_1d/*.csv</code> ; <code>output/gn_fp_2d/*.csv</code> ;<br><code>paper/gn_profile_impact.csv</code> ;<br><code>paper/gn_nmax_convergence.csv</code> ;<br><code>paper/gn_baseline_replacement.csv</code> ;<br><code>paper/gn_cardy_vs_phase_space.png</code> |
| Open-system diagnostics                                                            | <code>lindblad_chi_minimal.py</code> ;<br><code>extract_chi_open_system_*.py</code> ;<br><code>scan_chi_open_system_*.py</code>                                                                                                                                                               | <code>output/chi_open_system/*.csv</code> ;<br><code>paper/chi_open_system_sensitivity.csv</code>                                                                                                                                                                             |
| Other first-principles profiles                                                    | <code>extract_tcoh_*.py</code> ;<br><code>extract_eta_*.py</code> ;<br><code>extract_superrad_*.py</code> ;<br><code>scan_*.profile_impact.py</code>                                                                                                                                          | <code>paper/tcoh_profile_impact.csv</code> ;<br><code>paper/eta_profile_impact.csv</code> ;<br><code>paper/superrad_profile_impact.csv</code>                                                                                                                                 |
| Packaging and manifest                                                             | <code>package_repro_outputs.py</code> (wrapper stage)                                                                                                                                                                                                                                         | <code>repro/runs/&lt;RUN_ID&gt;/figures/</code> ;<br><code>repro/runs/&lt;RUN_ID&gt;/tables/</code> ;<br><code>artifact_index.csv</code> ; <code>checksums.sha256</code> ;<br><code>manifest.json</code>                                                                      |

3. `repro/latest/checksums.sha256` and `repro/latest/artifact_index.csv` are generated and non-empty.

## Appendix B: Action-Derived Effective Potential and WKB

### 1. Two-Center Harmonic Conformal Factor

We define the conformal factor by a projected Poisson problem on the static slice,

$$\nabla^2 \Omega(\mathbf{x}) = -4\pi a [\rho_\varepsilon(\mathbf{x} - \mathbf{x}_+) + \rho_\varepsilon(\mathbf{x} - \mathbf{x}_-)], \quad \Omega \rightarrow 1 \quad (|\mathbf{x}| \rightarrow \infty), \quad (\text{B1})$$

with  $\mathbf{x}_\pm = \pm(D/2)\hat{z}$ . Using the Green representation

$$\Omega(\mathbf{x}) = 1 + a \sum_{s=\pm} \int d^3x' \frac{\rho_\varepsilon(\mathbf{x}' - \mathbf{x}_s)}{|\mathbf{x} - \mathbf{x}'|}, \quad (\text{B2})$$

and the normalized Plummer kernel

$$\rho_\varepsilon(\mathbf{r}) = \frac{3\varepsilon^2}{4\pi(|\mathbf{r}|^2 + \varepsilon^2)^{5/2}}, \quad \int d^3r \rho_\varepsilon(\mathbf{r}) = 1, \quad (\text{B3})$$

the integral is analytic and gives

$$\Omega(\rho, z; D) = 1 + a \left( \frac{1}{\sqrt{\rho^2 + (z - D/2)^2 + \varepsilon^2}} + \frac{1}{\sqrt{\rho^2 + (z + D/2)^2 + \varepsilon^2}} \right) \quad (\text{B4})$$

where  $a$  is the geometric strength,  $\varepsilon$  is the core regularization scale, and  $D$  is the dual-center separation. Away from the cores ( $|\mathbf{x} - \mathbf{x}_\pm| \gg \varepsilon$ ), this satisfies  $\nabla^2 \Omega \approx 0$ . The radial identity

$$\nabla^2 \left( \frac{1}{\sqrt{r^2 + \varepsilon^2}} \right) = -\frac{3\varepsilon^2}{(r^2 + \varepsilon^2)^{5/2}} \quad (\text{B5})$$

gives the smeared Laplacian:

$$\nabla^2 \Omega = -3a\varepsilon^2 \left[ (r_+^2 + \varepsilon^2)^{-5/2} + (r_-^2 + \varepsilon^2)^{-5/2} \right] < 0 \quad (\text{B6})$$

which is equivalent to the source form in Eq. (6). This negative contribution is essential for forming the attractive wells in  $V_{\text{eff}}$ . As a reproducibility check, `verify_omega_geometric_origin.py` confirms  $\int d^3r \rho_\varepsilon = 1$  with absolute error  $3.75 \times 10^{-7}$  and validates the radial Laplacian identity with max relative error  $1.39 \times 10^{-4}$  in the non-asymptotic band (see `output/omega_geom_origin/`).

## 2. Veff Derivation from KG Equation

Starting from  $(\square_g - m_0^2 - \xi R)\Phi = 0$  with  $g_{\mu\nu} = \Omega^2 \eta_{\mu\nu}$ :

*a. Step 1:  $\square_g$  in conformal coordinates.*

$$\square_g \Phi = \Omega^{-2}(-\partial_t^2 + \nabla^2)\Phi + 2\Omega^{-3}\nabla\Omega \cdot \nabla\Phi \quad (\text{B7})$$

*b. Step 2: Time separation.* For  $\Phi = e^{-i\omega t}\phi(\mathbf{x})$ :

$$\nabla^2 \phi + 2\Omega^{-1}\nabla\Omega \cdot \nabla\phi + [\omega^2 - \Omega^2(m_0^2 + \xi R)]\phi = 0 \quad (\text{B8})$$

*c. Step 3: Conformal field rescaling.* Setting  $\phi = \Omega^{-1}\psi$  eliminates the first-derivative term. Using  $R = -6\Omega^{-3}\nabla^2\Omega$  for 4D conformally flat space:

$$\boxed{[-\nabla^2 + V_{\text{eff}}]\psi = \omega^2\psi, \quad V_{\text{eff}} = m_0^2\Omega^2 + (1 - 6\xi)\Omega^{-1}\nabla^2\Omega} \quad (\text{B9})$$

*d. Double-well formation condition.* For  $\xi < 1/6$  and  $\nabla^2\Omega < 0$  near cores, the curvature term contributes **negatively** to  $V_{\text{eff}}$ , creating attractive wells. The mass term  $m_0^2\Omega^2$  provides the continuum threshold at infinity.

*e. Self-consistency check.* At  $\xi = \xi_c = 1/6$  (conformal coupling):  $V_{\text{eff}} \rightarrow m_0^2\Omega^2$ , and the curvature contribution vanishes identically.  $\checkmark$

### 3. 1D On-Axis Reduction

For efficient scanning, we use the on-axis ( $\rho = 0$ ) reduction:

$$\left[-\frac{d^2}{dz^2} + U(z)\right]\psi_n(z) = E_n\psi_n(z), \quad U(z) = V_{\text{eff}}(\rho = 0, z) - m_0^2 \quad (\text{B10})$$

where  $E_n = \omega_n^2 - m_0^2$ . Bound states satisfy  $E_n < 0$ ; stable modes additionally require  $\omega_n^2 = m_0^2 + E_n > 0$ .

### 4. 2D Axisymmetric Validation

We validated the 1D reduction by comparing with the full 2D axisymmetric Laplacian:

$$\nabla^2\Omega = \frac{\partial^2\Omega}{\partial\rho^2} + \frac{1}{\rho}\frac{\partial\Omega}{\partial\rho} + \frac{\partial^2\Omega}{\partial z^2} \quad (\text{B11})$$

For the physical-gap parameters ( $a = 0.04$ ,  $\varepsilon = 0.1$ ,  $m_0 = 1$ ,  $\xi = 0.14$ ), the relative error at  $\rho = 0$  is:

| $D$ | Max $ U_{2\text{D}} - U_{1\text{D}} /\max U $ |
|-----|-----------------------------------------------|
| 6   | $4.0 \times 10^{-4}$                          |
| 12  | $4.0 \times 10^{-4}$                          |
| 18  | $4.0 \times 10^{-4}$                          |

The 1D 3D-identity approximation is accurate to  $< 0.1\%$ , validating its use for phase scans.

### 5. WKB Action Convention

The tunneling action through the central barrier is:

$$S_N = \int_{z_1}^{z_2} \sqrt{U(z) - E_N} dz \quad (\text{B12})$$

where  $z_{1,2}$  are the turning points satisfying  $U(z_{1,2}) = E_N$ . The tunneling suppression factor is:

$$r_N = \eta \cdot e^{-2S_N} \quad (\text{B13})$$

**Convention note:** For rates/probabilities we use  $e^{-2S}$ , while the level-splitting amplitude scales as  $e^{-S}$ . The empirical splitting relation below therefore tests the amplitude-level law.

## 6. Splitting–Action Consistency Check

The most stringent internal consistency check is the relation between level splitting and tunneling action. For symmetric double wells, WKB theory predicts  $\Delta E \propto e^{-S}$ . From our action-derived scan:

$$\boxed{\ln \Delta E \approx -1.01 S_1 + 0.69, \quad R^2 = 0.9999} \quad (\text{B14})$$

This near-perfect linear relation (Fig. 9) confirms that the **same**  $V_{\text{eff}}$  controls both the spectrum and the tunneling kinetics—not an engineered coincidence.

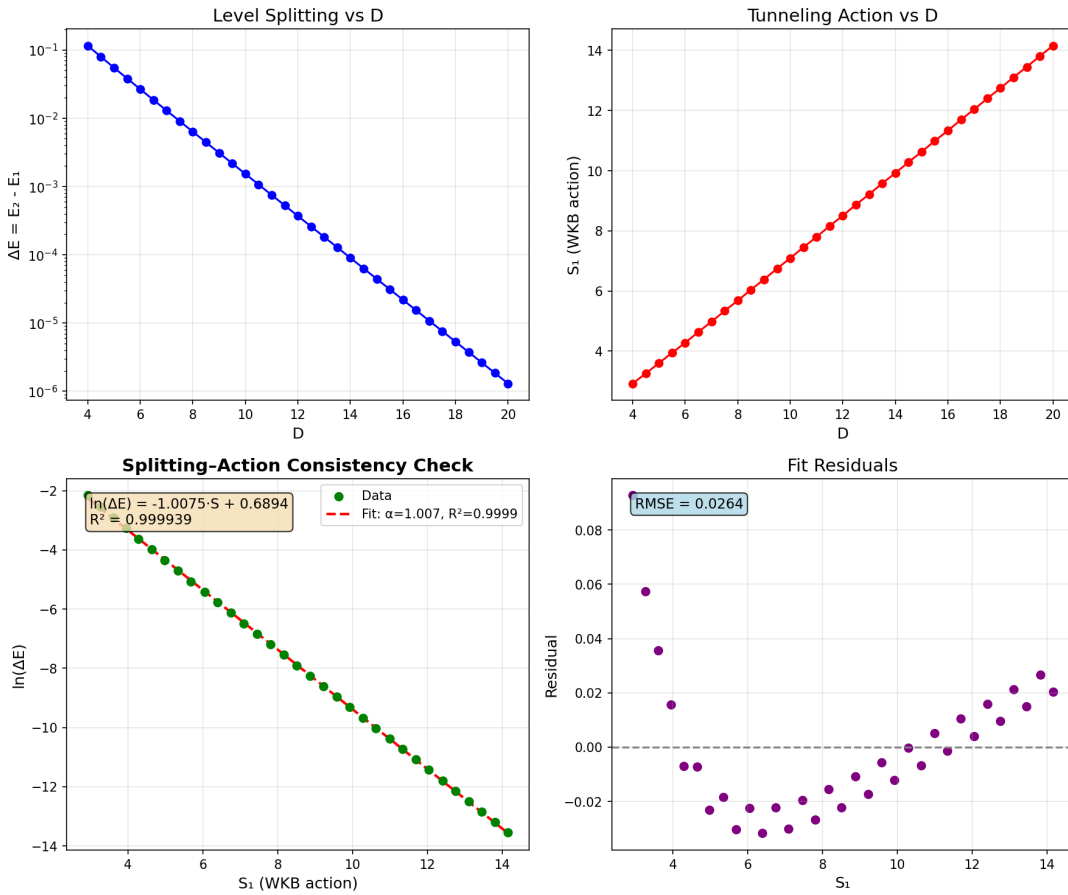


FIG. 9. Splitting–action consistency check. The level splitting  $\Delta E = E_2 - E_1$  follows  $\exp(-S_1)$  to  $R^2 = 0.9999$ , confirming the action-derived  $V_{\text{eff}}$  governs both spectrum and tunneling.

## 7. Action-Derived Numerical Results

Table VII shows the key spectral and tunneling quantities from the action-derived calculation. Grid convergence:  $|E_1|$  variation  $< 0.2\%$  for  $dz \in [0.01, 0.02]$  at fixed  $z_{\text{max}} = 80$ .



TABLE VII. Action-derived spectral and tunneling data. Parameters:  $a = 0.04$ ,  $\varepsilon = 0.1$ ,  $m_0 = 1$ ,  $\xi = 0.14$ .

| $D$ | $E_1$  | $E_2$  | $\omega_1$ | $S_1$ | $\Delta E$            |
|-----|--------|--------|------------|-------|-----------------------|
| 4   | -0.505 | -0.390 | 0.704      | 2.92  | $1.15 \times 10^{-1}$ |
| 8   | -0.476 | -0.469 | 0.724      | 5.68  | $6.32 \times 10^{-3}$ |
| 12  | -0.478 | -0.478 | 0.722      | 8.51  | $3.69 \times 10^{-4}$ |
| 16  | -0.481 | -0.481 | 0.721      | 11.33 | $2.19 \times 10^{-5}$ |
| 20  | -0.482 | -0.482 | 0.720      | 14.16 | $1.30 \times 10^{-6}$ |

### 8. Bound-State omega Convergence Benchmark

To make the surrogate-vs-exact distinction explicit, we report a direct bound-state benchmark for  $\omega_N$  from the same action-derived 1D operator chain used in Table VII. The settings are fixed to the physical-gap parameter set ( $a = 0.04$ ,  $\varepsilon = 0.1$ ,  $m_0 = 1$ ,  $\xi = 0.14$ ),  $z_{\max} = 80$ , and three resolutions: coarse ( $N_z = 4001$ ), mid ( $N_z = 6001$ ), fine ( $N_z = 8001$ ). The reproducible script is `code/extract_omega_exact_convergence.py`.

TABLE VIII. Bound-state  $\omega_N$  convergence benchmark at  $D = \{6, 12, 18\}$  from the action-derived operator chain. Relative errors are quoted versus fine grid.

| $D$ | level  | $N_z$ | $\omega_1$ | $\omega_2$ | $\Delta\omega_{12}$     | max rel. $\omega$ vs fine |
|-----|--------|-------|------------|------------|-------------------------|---------------------------|
| 6   | coarse | 4001  | 0.720883   | 0.738989   | $1.8107 \times 10^{-2}$ | $8.65 \times 10^{-4}$     |
| 6   | mid    | 6001  | 0.721345   | 0.739468   | $1.8123 \times 10^{-2}$ | $2.18 \times 10^{-4}$     |
| 6   | fine   | 8001  | 0.721500   | 0.739629   | $1.8129 \times 10^{-2}$ | 0                         |
| 12  | coarse | 4001  | 0.721781   | 0.722035   | $2.5438 \times 10^{-4}$ | $8.84 \times 10^{-4}$     |
| 12  | mid    | 6001  | 0.722258   | 0.722513   | $2.5533 \times 10^{-4}$ | $2.23 \times 10^{-4}$     |
| 12  | fine   | 8001  | 0.722418   | 0.722674   | $2.5565 \times 10^{-4}$ | 0                         |
| 18  | coarse | 4001  | 0.719404   | 0.719408   | $3.6700 \times 10^{-6}$ | $8.92 \times 10^{-4}$     |
| 18  | mid    | 6001  | 0.719885   | 0.719888   | $3.6943 \times 10^{-6}$ | $2.25 \times 10^{-4}$     |
| 18  | fine   | 8001  | 0.720046   | 0.720050   | $3.7025 \times 10^{-6}$ | 0                         |

This benchmark is the operator-level reference for  $\omega_N$  in the bound-state convention. In contrast, Appendix D uses generalized localized-extraction eigenvalues  $\lambda$  for mixing-channel extraction; the two quantities serve different roles and are not identified.

### 9. Fixed-dz Convergence

We verified numerical stability using fixed grid spacing  $dz$  (rather than fixed  $N_z$ ):

- For  $dz \in \{0.04, 0.02, 0.01\}$  and  $z_{\max} \in \{60, 80\}$ :  $E_1$  variation  $< 0.3\%$ ,  $S_1$  variation  $< 0.2\%$ .
- Results are independent of  $z_{\max}$  for  $z_{\max} > 60$  (domain truncation error negligible).

### Appendix C: First-Principles Symmetry-Channel Test for $\chi$

Using the explicit definition in Eq. (59), we computed  $M_{12}$  and  $\chi_N^{(\text{sym})}$  on 2D axisymmetric grids for  $D = \{6, 12, 18\}$  with coarse/mid/fine resolutions. The modes are normalized with the cylindrical measure  $2\pi\rho d\rho dz$ , and orthogonality satisfies  $|\langle\psi_1, \psi_2\rangle| < 10^{-15}$  in all runs.

TABLE IX. Fine-grid symmetry-channel extraction using Eq. (59). Parameters:  $a = 0.04$ ,  $\varepsilon = 0.1$ ,  $m_0 = 1$ ,  $\xi = 0.14$ .

| $D$ | $ M_{12}^{(\text{sym})} $ | $\chi_N^{(\text{sym})}$ | $ \langle\psi_1, \psi_2\rangle $ |
|-----|---------------------------|-------------------------|----------------------------------|
| 6   | $2.09 \times 10^{-16}$    | $1.66 \times 10^{-19}$  | $3.12 \times 10^{-17}$           |
| 12  | $6.03 \times 10^{-16}$    | $5.69 \times 10^{-19}$  | $3.47 \times 10^{-18}$           |
| 18  | $6.24 \times 10^{-16}$    | $6.20 \times 10^{-19}$  | $2.11 \times 10^{-16}$           |

These values are numerically consistent with symmetry-protected cancellation of the parity-symmetric overlap channel. Across coarse/mid/fine grids,  $M_{12}^{(\text{sym})}$  changes sign but remains at  $\mathcal{O}(10^{-16})$ , indicating no resolved nonzero mixing in this channel.

For this reason, the phenomenological Rank-2 coefficient in the main text is interpreted as a localized-channel effective coupling (Section VIII), not as  $\chi_N^{(\text{sym})}$  from Eq. (59).

### Appendix D: Localized-Channel First-Principles Extraction of $\chi$

We compute the nonzero mixing channel directly in localized basis using

$$M_{LR}^{(H)} = \frac{\lambda_2 - \lambda_1}{2}, \quad \chi_N^{(LR)} = \frac{|M_{LR}^{(H)}|}{\bar{\Gamma}_N}, \quad (\text{D1})$$

with  $\bar{\Gamma}_N$  from Eq. (33). Here  $\lambda_{1,2}$  are generalized operator eigenvalues from  $K\psi = \lambda M\psi$  in the localized extraction solver; they are not the bound-state energy convention  $E = \omega^2 - m_0^2 < 0$  used in Appendix B. The extraction uses `code/extract_chi_localized_2d.py` with fixed settings:

- Domain:  $\rho_{\max} = 3.0$ ,  $z_{\max} = D/2 + 6.0$ .
- Grids: coarse  $(d\rho, dz) = (0.12, 0.06)$ , mid  $(0.08, 0.04)$ , fine  $(0.06, 0.03)$ .
- Solver: generalized eigensystem (shift-invert low-mode targeting,  $\sigma = 2.5$ ), tolerance  $10^{-8}$ , maxiter  $3 \times 10^4$ .

TABLE X. Localized-channel extraction for  $D = \{6, 12, 18\}$  (coarse/mid/fine). Parameters:  $a = 0.04$ ,  $\varepsilon = 0.1$ ,  $m_0 = 1$ ,  $\xi = 0.14$ .

| $D$ | level  | $(d\rho, dz)$ | $(N_\rho, N_z)$ | $\lambda_1$ | $\lambda_2$ | $M_{LR}^{(H)}$           | $\chi_N^{(LR)}$          | $ M_{12}^{(\text{sym})} $ |
|-----|--------|---------------|-----------------|-------------|-------------|--------------------------|--------------------------|---------------------------|
| 6   | coarse | (0.12, 0.06)  | (25, 300)       | 2.14765     | 2.60073     | $2.26541 \times 10^{-1}$ | $4.13261 \times 10^{-4}$ | $5.29 \times 10^{-17}$    |
| 6   | mid    | (0.08, 0.04)  | (38, 450)       | 2.14362     | 2.59798     | $2.27178 \times 10^{-1}$ | $4.17350 \times 10^{-4}$ | $2.44 \times 10^{-19}$    |
| 6   | fine   | (0.06, 0.03)  | (50, 600)       | 2.16611     | 2.62106     | $2.27474 \times 10^{-1}$ | $4.01827 \times 10^{-4}$ | $1.56 \times 10^{-18}$    |
| 12  | coarse | (0.12, 0.06)  | (25, 400)       | 2.36032     | 2.71730     | $1.78492 \times 10^{-1}$ | $2.27261 \times 10^{-4}$ | $2.89 \times 10^{-17}$    |
| 12  | mid    | (0.08, 0.04)  | (38, 600)       | 2.35598     | 2.71374     | $1.78883 \times 10^{-1}$ | $2.29384 \times 10^{-4}$ | $6.96 \times 10^{-17}$    |
| 12  | fine   | (0.06, 0.03)  | (50, 800)       | 2.37821     | 2.73632     | $1.79054 \times 10^{-1}$ | $2.21414 \times 10^{-4}$ | $1.03 \times 10^{-17}$    |
| 18  | coarse | (0.12, 0.06)  | (25, 500)       | 2.22204     | 2.49466     | $1.36311 \times 10^{-1}$ | $2.18683 \times 10^{-4}$ | $1.94 \times 10^{-17}$    |
| 18  | mid    | (0.08, 0.04)  | (38, 750)       | 2.21686     | 2.49001     | $1.36575 \times 10^{-1}$ | $2.21053 \times 10^{-4}$ | $1.54 \times 10^{-15}$    |
| 18  | fine   | (0.06, 0.03)  | (50, 1000)      | 2.23864     | 2.51202     | $1.36694 \times 10^{-1}$ | $2.13187 \times 10^{-4}$ | $2.19 \times 10^{-16}$    |

- Acceptance criteria:  $\delta_{\Delta E} \equiv |\Delta E_{\text{level}} - \Delta E_{\text{fine}}|/|\Delta E_{\text{fine}}| < 5\%$ ,  $\delta_\chi \equiv |\chi_{\text{level}} - \chi_{\text{fine}}|/|\chi_{\text{fine}}| < 5\%$ , and null-channel absolute bound  $|M_{12}^{(\text{sym})}| < 10^{-12}$ .

The complete pipeline, scripts, and raw tables used here are available in the project repository: <https://github.com/boypatrick/PSLT>.

Across all non-fine runs in Table X, we obtain

$$\max \delta_{\Delta E} = 4.10 \times 10^{-3}, \quad \max \delta_\chi = 3.86 \times 10^{-2}, \quad \max |M_{12}^{(\text{sym})}| = 1.54 \times 10^{-15}, \quad (\text{D2})$$

which satisfies the stated convergence and absolute-null criteria.

For interpolation support and reproducibility of scan-level diagnostics, we additionally ran a fine-grid-only auxiliary sweep on integer separations  $D = 4, \dots, 20$  using the same solver settings (`--full-scan` in `extract_chi_localized_2d.py`). These points are used as profile-support data in the repository, while the formal acceptance criteria above remain defined by the benchmark convergence set  $D = \{6, 12, 18\}$ .

Figure 10 shows the fine-grid 2D parity eigenmodes used in the localized extraction at  $D = \{6, 12, 18\}$ . The expected even/odd nodal structure is manifest and remains stable across the three benchmark separations.

To test profile uncertainty in the global map, we benchmarked six  $\chi_N^{(LR)}(D)$  choices built from the same fine-grid knots  $(D, \chi_N^{(LR)}) = \{(6, 4.02 \times 10^{-4}), (12, 2.21 \times 10^{-4}), (18, 2.13 \times 10^{-4})\}$ : linear interpolation, log-linear exponential fit  $\ln \chi = -7.5956 - 0.05282 D$ , and  $\pm 20\%$  amplitude rescalings of each profile. Re-running the full  $60 \times 60$  scan keeps the generation fractions fixed and yields a narrow  $H \rightarrow \mu\mu$  acceptance spread:

$$f(\mathcal{R}_3 > 0.90) = 0.932, \quad f(\mathcal{R}_3 > 0.95) = 0.217, \quad f(\chi_{\mu\mu}^2 < 4) \in [0.1625, 0.1681]. \quad (\text{D3})$$

TABLE XI. Auxiliary fine-grid localized profile points from the full-scan run ( $D = 4\text{--}20$ ). Source CSV: `output/chi_fp_2d/localized_chi_D4-5-...-20.csv`.

| $D$ | $\lambda_1$ | $\lambda_2$ | $\chi_N^{(LR)}$        |
|-----|-------------|-------------|------------------------|
| 4   | 2.06671     | 2.56696     | $5.266 \times 10^{-4}$ |
| 5   | 2.34635     | 2.85471     | $3.311 \times 10^{-4}$ |
| 6   | 2.16611     | 2.62106     | $4.018 \times 10^{-4}$ |
| 7   | 2.42273     | 2.88524     | $2.661 \times 10^{-4}$ |
| 8   | 2.24778     | 2.66479     | $3.202 \times 10^{-4}$ |
| 9   | 2.10111     | 2.48071     | $3.758 \times 10^{-4}$ |
| 10  | 2.32055     | 2.70644     | $2.623 \times 10^{-4}$ |
| 11  | 2.17620     | 2.52926     | $3.064 \times 10^{-4}$ |
| 12  | 2.37821     | 2.73632     | $2.214 \times 10^{-4}$ |
| 13  | 2.24440     | 2.57510     | $2.554 \times 10^{-4}$ |
| 14  | 2.42772     | 2.76173     | $1.906 \times 10^{-4}$ |
| 15  | 2.30010     | 2.61030     | $2.181 \times 10^{-4}$ |
| 16  | 2.47457     | 2.78817     | $1.661 \times 10^{-4}$ |
| 17  | 2.34884     | 2.64089     | $1.895 \times 10^{-4}$ |
| 18  | 2.23864     | 2.51202     | $2.132 \times 10^{-4}$ |
| 19  | 2.39519     | 2.67161     | $1.663 \times 10^{-4}$ |
| 20  | 2.28581     | 2.54509     | $1.867 \times 10^{-4}$ |

TABLE XII. Extended profile-sensitivity test for  $\chi_N^{(LR)}(D)$  in the  $60 \times 60$  global scan. Source data file: `paper/chi_profile_robustness.csv`.

| Profile              | $f(\mathcal{R}_3 > 0.90)$ | $f(\mathcal{R}_3 > 0.95)$ | $f(\chi_{\mu\mu}^2 < 4)$ | Best $(D, \eta)$ |
|----------------------|---------------------------|---------------------------|--------------------------|------------------|
| Linear interpolation | 0.932                     | 0.217                     | 0.168                    | (9.42, 1.29)     |
| Exponential fit      | 0.932                     | 0.217                     | 0.163                    | (9.15, 1.94)     |
| Linear $\times 0.8$  | 0.932                     | 0.217                     | 0.167                    | (8.61, 3.03)     |
| Linear $\times 1.2$  | 0.932                     | 0.217                     | 0.168                    | (8.07, 2.26)     |
| Exp-fit $\times 0.8$ | 0.932                     | 0.217                     | 0.163                    | (8.61, 3.10)     |
| Exp-fit $\times 1.2$ | 0.932                     | 0.217                     | 0.163                    | (8.88, 2.13)     |

The best-fit point moves within  $(D, \eta) \approx (8.07\text{--}9.42, 1.29\text{--}3.10)$  across these six profiles. Thus, at current scan resolution, profile-interpolation uncertainty is subdominant for  $\mathcal{R}_3$  and controlled for the  $H \rightarrow \mu\mu$  acceptance metric.

As a stronger stress test, we further rescaled the localized profile by large factors and tracked boundary movement directly on the  $(D, \eta)$  grid. Figure 11 shows that, at the present  $60 \times 60$  scan resolution, the  $\mathcal{R}_3 > 0.90$  mask is stable up to  $\times 10^4$  and first shifts at  $\times 10^5$ , while the

Appendix C: 2D Axisymmetric Eigenmodes for Localized-Channel Extraction  
 Fine grids with  $(d\rho, dz) = (0.06, 0.03)$  for  $D = \{6, 12, 18\}$

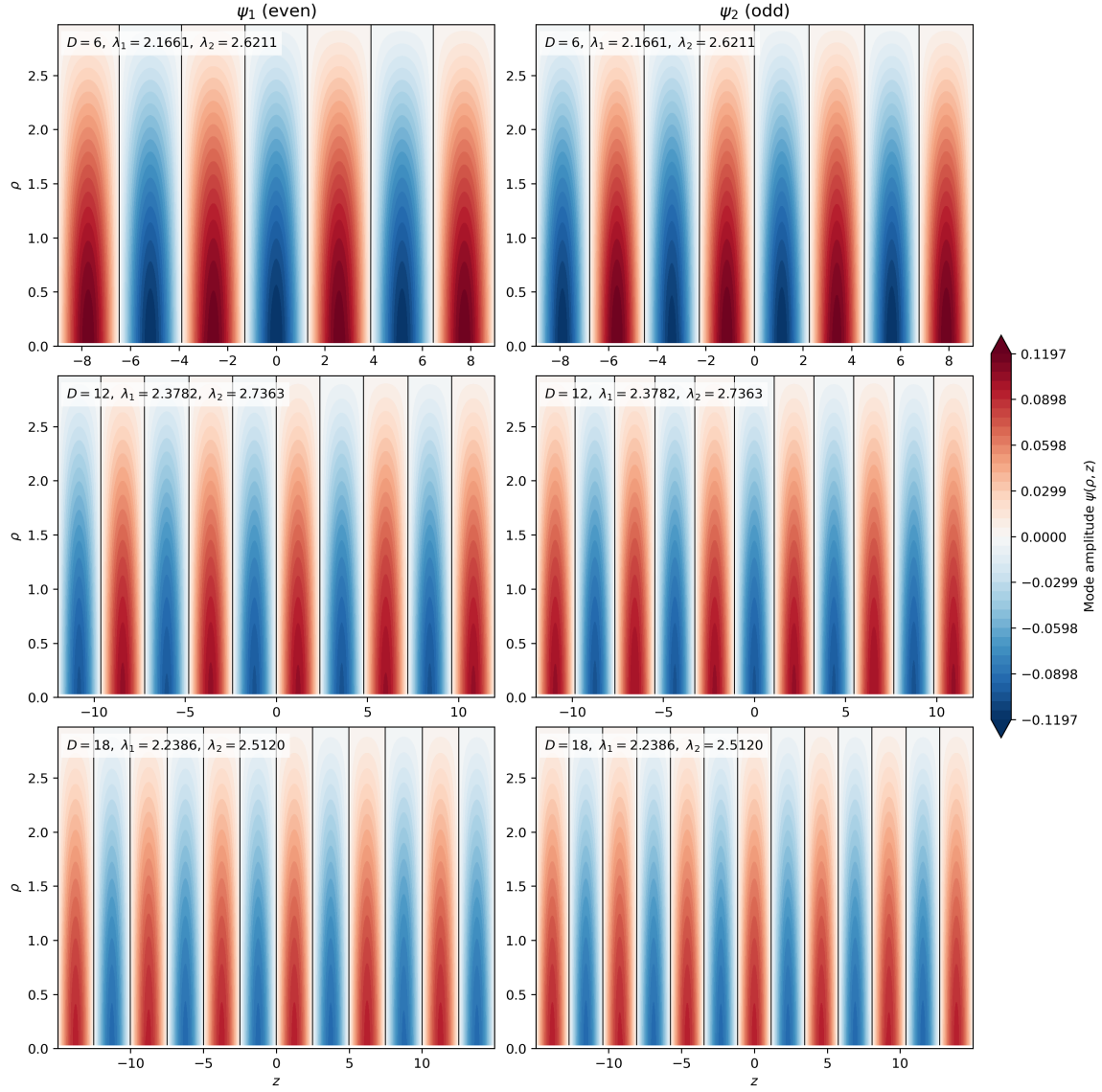


FIG. 10. Fine-grid 2D axisymmetric eigenmodes used in Appendix D. Each row corresponds to one benchmark separation ( $D = 6, 12, 18$ ); columns show the lowest even/odd parity modes ( $\psi_1, \psi_2$ ) entering the localized-channel extraction pipeline.

$H \rightarrow \mu\mu$  acceptance mask already shifts for the user-requested  $\times 0.5$  and  $\times 2$  cases (changed-cell fractions 0.0331 and 0.0258, respectively).

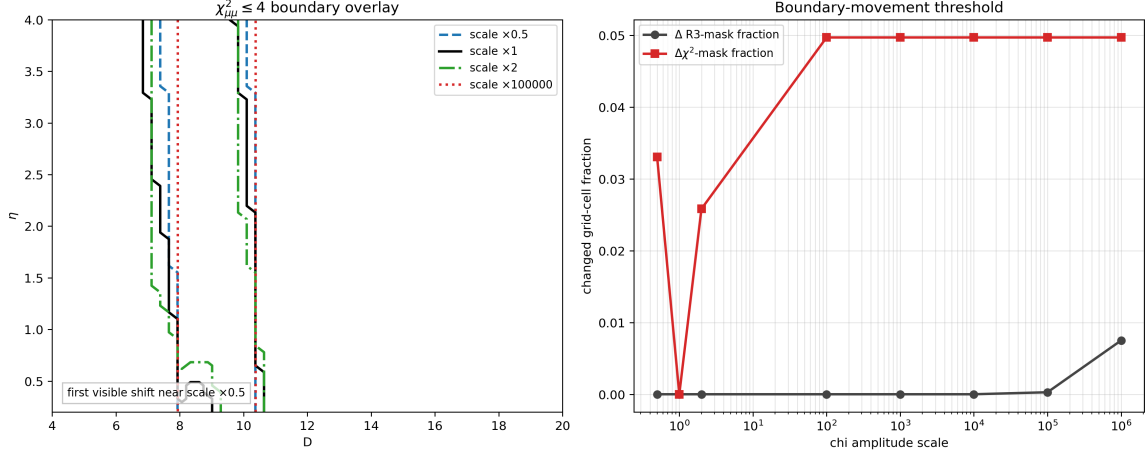


FIG. 11. Strong  $\chi$ -amplitude stress test on the global map. Left: overlay of  $\chi^2_{\mu\mu} \leq 4$  boundary for selected scales, including the requested  $\times 0.5$  and  $\times 2$ . Right: changed-cell fractions versus scale (relative to baseline  $\times 1$ ), showing early movement for the  $H \rightarrow \mu\mu$  mask and delayed movement for the  $\mathcal{R}_3$  mask. Source data file: `output/chi_fp_2d/chi_scale_stress_test.csv`.

TABLE XIII. Preliminary 1D phase-space candidate  $g_N^{(\text{ps})}$  at  $D = 12$ . Source CSV: `output/gn_fp_1d/gn_phase_space_candidate_D12.csv`.

| level  | $dz$ | $E_1$     | $E_2$     | $E_3$     | $g_1^{(\text{ps})}$ | $g_2^{(\text{ps})}$ | max rel. $g$ vs fine  |
|--------|------|-----------|-----------|-----------|---------------------|---------------------|-----------------------|
| coarse | 0.04 | -0.636696 | -0.608074 | -0.575010 | 1.056725            | 1.054421            | $1.31 \times 10^{-2}$ |
| mid    | 0.02 | -0.637061 | -0.608419 | -0.575335 | 1.066813            | 1.074099            | $5.30 \times 10^{-3}$ |
| fine   | 0.01 | -0.637233 | -0.608582 | -0.575488 | 1.063919            | 1.068433            | 0                     |

## Appendix E: Preliminary First-Principles Replacement Candidate Checks

### 1. 1D Phase-Space Candidate for $g_N$

To test a first-principles replacement direction for Eq. (18), we evaluated the semiclassical 1D candidate on the same action-derived axis potential:

$$\rho_{\text{WKB}}(E; D) = \frac{1}{2\pi} \int_{U(z; D) < E} \frac{dz}{\sqrt{E - U(z; D)}}, \quad (\text{E1})$$

$$g_N^{(\text{ps})}(D) = 1 + \int_{E_{\min}(D)}^{E_N(D)} \rho_{\text{WKB}}(E'; D) dE', \quad E_{\min}(D) \equiv \min_z U(z; D). \quad (\text{E2})$$

Using the reproducible script `extract_gn_phase_space_candidate.py` (in the `code/` directory), we ran a coarse/mid/fine test at  $D = 12$  (Table XIII). The candidate is numerically stable at the few- $10^{-3}$  level versus fine grid.

TABLE XIV. Fine-grid 2D phase-space profile candidate at  $D = \{6, 12, 18\}$ . Source CSV: `output/gn_fp_2d/gn_phase_space_2d_D6-12-18.csv`.

| $D$ | $\lambda_1$ | $\lambda_2$ | $\lambda_3$ | $\hat{g}_1$            | $\hat{g}_2$            |
|-----|-------------|-------------|-------------|------------------------|------------------------|
| 6   | 0.719476    | 0.804126    | 0.945018    | $9.867 \times 10^{-2}$ | $4.370 \times 10^{-1}$ |
| 12  | 0.686186    | 0.747510    | 0.822830    | $1.193 \times 10^{-1}$ | $5.146 \times 10^{-1}$ |
| 18  | 0.667848    | 0.713807    | 0.767832    | $1.291 \times 10^{-1}$ | $5.294 \times 10^{-1}$ |

## 2. 2D Axisymmetric Benchmark for $g_N^{(\text{ps})}$

To reduce projection ambiguity, we also ran a 2D axisymmetric benchmark on  $D = \{6, 12, 18\}$  using the same localized extraction chain as Appendix D: identical generalized operator build, identical box/range convention, and the same shift-invert low-mode settings ( $\sigma = 2.5$ ,  $\text{tol} = 10^{-8}$ ,  $\text{maxiter} = 3 \times 10^4$ ). We define

$$N_{\text{ps}}(E) = \frac{1}{4\pi} \int d^2x d^2p \Theta(E - U - p^2) = \frac{1}{2} \int \rho d\rho dz [E - U(\rho, z)]_+, \quad (\text{E3})$$

$$\mathcal{W}_N \equiv [\lambda_1, \lambda_N], \quad g_{N, \text{raw}}^{(\text{ps})} = 1 + \int_{\mathcal{W}_N} \rho_{\text{ps}}(E') dE' = 1 + N_{\text{ps}}(\lambda_N) - N_{\text{ps}}(\lambda_1), \quad \hat{g}_N \equiv \frac{g_{N, \text{raw}}^{(\text{ps})}}{g_{3, \text{raw}}^{(\text{ps})}}. \quad (\text{E4})$$

The script `extract_gn_phase_space_2d.py` (in `code/`) exports both the low- $N$  profile table and the low-mode spectral table with explicit window bounds (source CSVs: `output/gn_fp_2d/gn_phase_space_2d_D6-12-18.csv`, `output/gn_fp_2d/gn_phase_space_2d_spectrum_D6-12-18.csv`). For non-fine levels, the maximum profile deviation is

$$\max_{\text{non-fine}} \left( \max_N \frac{|\hat{g}_N - \hat{g}_N^{(\text{fine})}|}{|\hat{g}_N^{(\text{fine})}|} \right) = 2.35 \times 10^{-2}. \quad (\text{E5})$$

As in the 1D candidate check, this benchmark is reported as a replacement candidate. To test map-level impact, we connected both candidates to the main scan through `g_mode` choices (`fp_2d_full`, `cardy`, `fp_1d`, `fp_2d`) and the reproducible script `scan_gn_profile_impact.py`. Results in Table XV show that the main conclusion is retained: relative to baseline `fp_2d_full`, the largest absolute shift is 0.0669 in  $f(\mathcal{R}_3 > 0.90)$  (for `fp_2d`), while high- $N$  runaway remains controlled with  $f(N_{\text{win}} > 3) \simeq 2.78 \times 10^{-4}$  in all tested modes.

Across  $N_{\text{max}} = 20, 30, 40$ , both baseline and `fp_2d` scenarios are numerically stable at map level: the key fractions  $f(\mathcal{R}_3 > 0.90)$ ,  $f(\chi_{\mu\mu}^2 < 4)$ , and  $f(N_{\text{win}} > 3)$  remain unchanged in this scan.

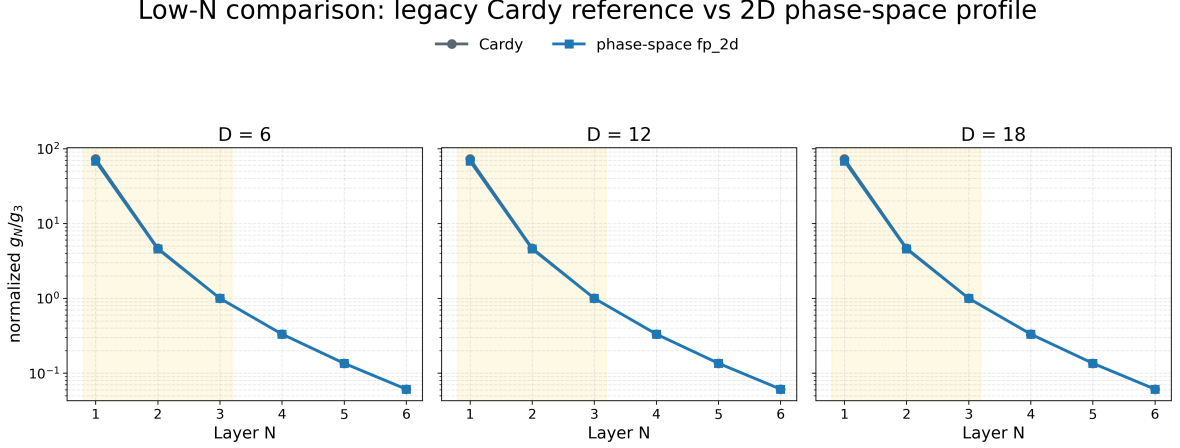


FIG. 12. Low- $N$  comparison between legacy Cardy reference and 2D phase-space profile on  $D = \{6, 12, 18\}$ , shown as normalized ratios  $g_N/g_3$  (log scale). The highlighted band marks  $N = 1, 2, 3$ . Source files: `paper/gn_cardy_vs_phase_space.png`, `paper/gn_cardy_vs_phase_space.csv`.

TABLE XV. Map-level impact of first-principles  $g_N$  profiles on the  $60 \times 60$  scan (baseline is `fp_2d_full`; legacy Cardy retained as comparator). Source CSV: `paper/gn_profile_impact.csv`.

| case                             | $f(\mathcal{R}_3 > 0.90)$ | $f(\chi_{\mu\mu}^2 < 4)$ | $f(N_{\text{win}} > 3)$ | mean tail prob. |
|----------------------------------|---------------------------|--------------------------|-------------------------|-----------------|
| baseline <code>fp_2d_full</code> | 0.927                     | 0.1614                   | 0.0003                  | 0.1267          |
| legacy Cardy reference           | 0.877                     | 0.1592                   | 0.0003                  | 0.1125          |
| $g_N$ fp_1d profile              | 0.860                     | 0.1597                   | 0.0003                  | 0.1137          |
| $g_N$ fp_2d profile              | 0.860                     | 0.1597                   | 0.0003                  | 0.1144          |

### 3. Baseline Cross-Check Under $g\_mode = \text{fp\_2d\_full}$

To implement a strict same-framework upgrade, we set `g_mode=fp_2d_full` as the baseline. Instead of direct low- $N$  substitution plus geometric continuation, this mode uses an explicit bounded microcanonical window for  $N \leq 3$  and an energy-gap tail prescription for  $N > 3$ :

$$g_N^{(\text{full})}(D) = g_3^{(\text{ps})}(D) \hat{g}_N(D), \quad \hat{g}_n = \left( \hat{g}_n^{(\text{dir})} \right)^{1-\alpha} \left( \hat{g}_n^{(\text{win})} \right)^\alpha \quad (n = 1, 2, 3), \quad (\text{E6})$$

with

$$\hat{g}_1^{(\text{win})} = 1 + N_{\text{ps}}(\lambda_3) - N_{\text{ps}}(\lambda_1), \quad \hat{g}_2^{(\text{win})} = 1 + N_{\text{ps}}(\lambda_3) - N_{\text{ps}}(\lambda_2), \quad \hat{g}_3^{(\text{win})} = 1, \quad (\text{E7})$$

with  $g_3^{(\text{ps})}(D) \equiv 1 + N_{\text{ps}}(\lambda_3) - N_{\text{ps}}(\lambda_1)$  from the same 2D spectrum table. For  $N > 3$ ,

$$\hat{g}_N^{(\text{target})} = \left( \frac{\Delta N_{\text{ps}}^{(N)}}{\Delta N_{\text{ps}}^{(3)}} \right)^s \exp \left[ -\beta \frac{\lambda_N - \lambda_3}{\lambda_3 - \lambda_2} \right], \quad (\text{E8})$$

followed by the same per-step finite-volume clipping used in `pslt_lib.py`. In the present baseline run, we use  $(\alpha, \beta, s) = (0.8, 1.1, 0.0)$ . Results are summarized in Table XVII (source CSV:



TABLE XVI.  $N_{\max}$  convergence check for baseline and  $g_N$  fp\_2d map-level summaries. Source CSV: `paper/gn_nmax_convergence.csv`.

| case                | $N_{\max}$ | $f(\mathcal{R}_3 > 0.90)$ | $f(\chi_{\mu\mu}^2 < 4)$ | $f(N_{\text{win}} > 3)$ | mean tail prob. |
|---------------------|------------|---------------------------|--------------------------|-------------------------|-----------------|
| baseline fp_2d_full | 20         | 0.927                     | 0.1614                   | 0.0003                  | 0.126709        |
| baseline fp_2d_full | 30         | 0.927                     | 0.1614                   | 0.0003                  | 0.126709        |
| baseline fp_2d_full | 40         | 0.927                     | 0.1614                   | 0.0003                  | 0.126709        |
| $g_N$ fp_2d profile | 20         | 0.860                     | 0.1597                   | 0.0003                  | 0.114352        |
| $g_N$ fp_2d profile | 30         | 0.860                     | 0.1597                   | 0.0003                  | 0.114352        |
| $g_N$ fp_2d profile | 40         | 0.860                     | 0.1597                   | 0.0003                  | 0.114352        |

TABLE XVII. Map-level baseline cross-check for  $g_N$ : `fp_2d_full` baseline versus legacy Cardy reference on the same  $60 \times 60$  grid. Source CSV: `paper/gn_baseline_replacement.csv`.

| case                   | $f(\mathcal{R}_3 > 0.90)$ | $f(\chi_{\mu\mu}^2 < 4)$ | $f(N_{\text{win}} > 3)$ | $\Delta f(\mathcal{R}_3 > 0.90)$ vs baseline |
|------------------------|---------------------------|--------------------------|-------------------------|----------------------------------------------|
| baseline fp_2d_full    | 0.9269                    | 0.1614                   | 0.0003                  | 0.0000                                       |
| legacy Cardy reference | 0.8769                    | 0.1592                   | 0.0003                  | -0.0500                                      |

`paper/gn_baseline_replacement.csv`). Relative to baseline `fp_2d_full`, the legacy Cardy reference lowers  $f(\mathcal{R}_3 > 0.90)$  by 0.0500 (from 0.9269 to 0.8769), lowers  $f(\chi_{\mu\mu}^2 < 4)$  by 0.0022 (from 0.1614 to 0.1592), and keeps  $f(N_{\text{win}} > 3) \simeq 2.78 \times 10^{-4}$ . Therefore, the map-level main conclusion is retained under this baseline switch.

Across  $N_{\max} = 20, 30, 40$ , both rows in Table XVII are numerically stable in this setup (source CSV: `paper/gn_baseline_replacement_nmax.csv`).

#### 4. Geometry-Informed Lindblad Scan-Ready Module for Open-System $\chi$

As a methodological bridge (not a replacement of Appendix D), we consider a two-level open-system model in the standard GKSL/Lindblad form [10, 11]:

$$\dot{\rho} = -i[H, \rho] + \gamma_{\phi} \mathcal{D}[\sigma_z] \rho + \gamma_{\text{mix}} \mathcal{D}[\sigma_x] \rho, \quad H = \begin{pmatrix} 0 & \Delta/2 \\ \Delta/2 & 0 \end{pmatrix}. \quad (\text{E9})$$

As external motivation for this diagnostic direction, recent entanglement-based flavor analyses report that minimizing scattering-generated entanglement can qualitatively reproduce observed CKM/PMNS patterns [12]. In this manuscript we use that connection only as qualitative support for coherence-sensitive diagnostics, not as a baseline closure assumption. For this module we set  $\gamma_{\text{mix}}(D) \equiv M_{LR}^{(H)}(D)$  from Appendix D, and define a geometry-informed dephasing proxy

$$\gamma_{\phi}(D) \equiv \sqrt{\left\langle (\delta V)^2 \right\rangle_{\rho}}, \quad \delta V = V_{\text{eff}} - \bar{V}_{\text{eff}}(r), \quad (\text{E10})$$

TABLE XVIII. Geometry-informed open-system demonstrator at  $D = \{6, 12, 18\}$ . Source CSV: `output/chi_open_system/chi_open_system_geometry_D6-12-18.csv`.

| $D$ | $\gamma_\phi$ | $\gamma_{\text{mix}}$ | $C_{\text{max}}$ | $\chi_{\text{eff}}^{(\text{proxy})}$ | $\chi_N^{(LR)}$       | ratio |
|-----|---------------|-----------------------|------------------|--------------------------------------|-----------------------|-------|
| 6   | 0.656901      | 0.227474              | 0.077658         | $6.24 \times 10^{-5}$                | $4.02 \times 10^{-4}$ | 0.155 |
| 12  | 0.744389      | 0.179054              | 0.063765         | $2.82 \times 10^{-5}$                | $2.21 \times 10^{-4}$ | 0.128 |
| 18  | 0.805854      | 0.136694              | 0.051381         | $2.19 \times 10^{-5}$                | $2.13 \times 10^{-4}$ | 0.103 |

TABLE XIX. Open-system sensitivity scan with profiled  $(\gamma_\phi, \gamma_{\text{mix}})$  on the  $60 \times 60$  map. Source CSV: `paper/chi_open_system_sensitivity.csv`.

| case                               | ratio min | ratio max | ratio mean |
|------------------------------------|-----------|-----------|------------|
| baseline localized                 | 1.000     | 1.000     | 1.000      |
| open-system base (1, 1)            | 0.085     | 0.213     | 0.127      |
| $\phi \times 0.5$ , mix $\times 1$ | 0.146     | 0.216     | 0.178      |
| $\phi \times 2$ , mix $\times 1$   | 0.059     | 0.115     | 0.082      |
| $\phi \times 1$ , mix $\times 0.5$ | 0.058     | 0.109     | 0.079      |
| $\phi \times 1$ , mix $\times 2$   | 0.148     | 0.228     | 0.184      |
| $\phi \times 0.5$ , mix $\times 2$ | 0.206     | 0.275     | 0.232      |
| $\phi \times 2$ , mix $\times 0.5$ | 0.032     | 0.068     | 0.046      |

with cylindrical weighting  $\langle \cdots \rangle_\rho$ . The minimal Lindblad evolution is generated by `lindblad_chi_minimal.py`, and geometry-profile extraction is handled by `extract_chi_open_system_geometry.py`. These scripts output reproducible diagnostics such as

$$C_{\text{max}} \equiv \max_t |\rho_{LR}(t)|, \quad P_{\text{max}} \equiv \max_t \rho_{RR}(t), \quad (\text{E11})$$

and a normalized proxy  $\chi_{\text{eff}}^{(\text{proxy})} = 2\gamma_{\text{mix}}C_{\text{max}}/\bar{\Gamma}_N$ . Results for  $D = \{6, 12, 18\}$  are summarized in Table XVIII. Extending the same profile to  $D = 4, \dots, 20$  gives a stable baseline ratio band  $\chi_{\text{open}}/\chi_{LR} \in [0.0852, 0.2127]$  with mean 0.1269 (source CSV: `output/chi_open_system/chi_open_ratio.band`).

Across all sensitivity cases in Table XIX, the generation metrics remain unchanged on this grid ( $f(\mathcal{R}_3 > 0.90) = 0.932$ ,  $f(\mathcal{R}_3 > 0.95) = 0.217$ ,  $f(N_{\text{win}} > 3) \simeq 2.78 \times 10^{-4}$ ), while the  $H \rightarrow \mu\mu$  acceptance fraction shifts within a controlled band  $f(\chi_{\mu\mu}^2 < 4) \in [0.1500, 0.1558]$  around the localized reference. These values are model-dependent because the Lindblad structure is still an effective environment ansatz and not yet a microscopic EYMH bath derivation; therefore, `chi_mode=open_system` is treated as scan-ready diagnostic only and is not propagated into baseline reported figures.

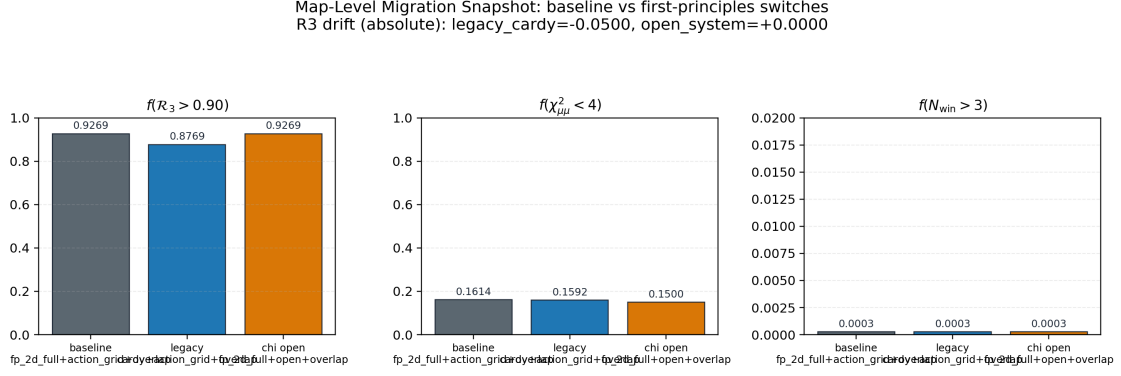


FIG. 13. Map-level migration snapshot across three scenarios. Left to right:  $f(\mathcal{R}_3 > 0.90)$ ,  $f(\chi^2_{\mu\mu} < 4)$ , and  $f(N_{\text{win}} > 3)$ .

TABLE XX. Cross-module migration comparison from `paper/first_principles_migration_summary.csv`.

| scenario                                    | $f(\mathcal{R}_3 > 0.90)$ | $\Delta f(\mathcal{R}_3 > 0.90)$ | $f(\chi^2_{\mu\mu} < 4)$ | $f(N_{\text{win}} > 3)$ |
|---------------------------------------------|---------------------------|----------------------------------|--------------------------|-------------------------|
| baseline (fp_2d.full + localized_grid)      | 0.927                     | 0.0000                           | 0.1614                   | 0.0003                  |
| legacy reference (cardy + localized_grid)   | 0.877                     | -0.0500                          | 0.1592                   | 0.0003                  |
| $\chi$ migration (fp_2d.full + open_system) | 0.927                     | 0.0000                           | 0.1500                   | 0.0003                  |

## 5. Cross-Module Migration Snapshot

To compare the two main migration directions on the same map-level metrics under the updated baseline, we aggregate three scenarios: baseline (`fp_2d.full + localized_grid`), legacy reference (`cardy + localized_grid`), and open-system  $\chi$  (`fp_2d.full + open_system`). The reproducible script is `scan_first_principles_migration_summary.py`; outputs are `paper/first_principles_migration_summary.csv` and Fig. 13.

## 6. Reference-Anchor Sensitivity and Dynamic Selection Rule

Because Eq. (51) is normalized to a reference point  $(D_0, \eta_0)$ , we explicitly audit anchor dependence with two reproducible scripts: `select_reference_anchor.py` and `scan_reference_anchor_sensitivity.py`. The workflow is:

1. build a fixed-reference baseline map at  $(D_0, \eta_0) = (10, 1)$  on the same  $60 \times 60$  grid;
2. extract three candidate anchors: fixed-grid point,  $\chi^2$ -best [Eq. (52)], and robust-center [Eq. (53)];
3. re-anchor Eq. (51) with each candidate and recompute map summaries.

TABLE XXI. Reference-anchor candidates and re-anchored map metrics. Source CSV: `paper/reference_anchor_candidates.csv`; selector JSON: `paper/reference_anchor_choice.json`.

| mode           | $(D_0, \eta_0)$ | $\chi^2$ at selector point | $f(\chi_{\mu\mu}^2 < 4)$ (re-anchored) | best $\chi^2$ (re-anchored) |
|----------------|-----------------|----------------------------|----------------------------------------|-----------------------------|
| fixed-grid     | (9.97, 0.97)    | $8.32 \times 10^{-1}$      | 0.1611                                 | $3.06 \times 10^{-6}$       |
| $\chi^2$ -best | (8.07, 2.71)    | $6.39 \times 10^{-5}$      | 0.1028                                 | $7.47 \times 10^{-6}$       |
| robust-center  | (8.61, 3.29)    | $8.55 \times 10^{-1}$      | 0.1189                                 | $1.67 \times 10^{-5}$       |

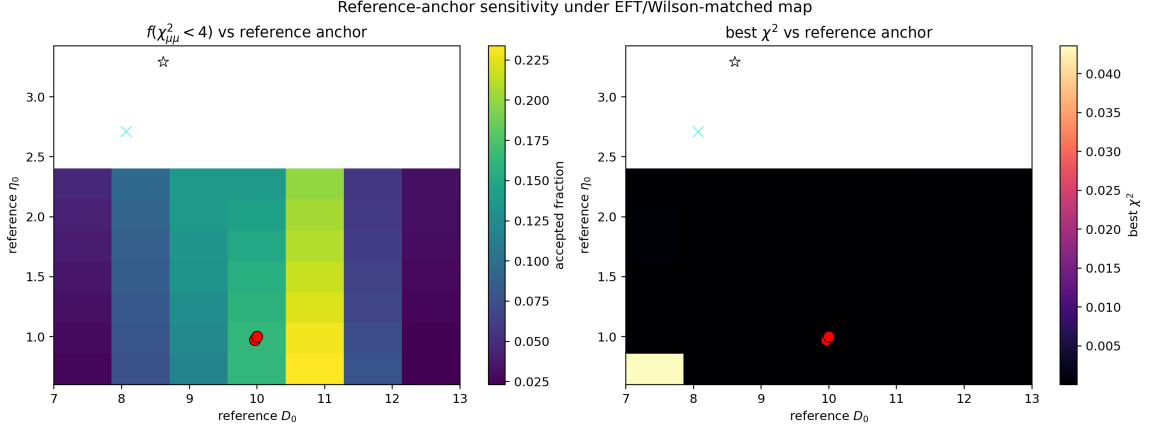


FIG. 14. Systematic sensitivity over reference anchors  $(D_0, \eta_0)$  under the EFT/Wilson-matched map. Left:  $f(\chi_{\mu\mu}^2 < 4)$ ; right: best  $\chi^2$ . Overlaid markers indicate baseline fixed,  $\chi^2$ -best, and robust-center selectors. Source files: `paper/reference_anchor_sensitivity.csv`, `paper/reference_anchor_sensitivity.png`.

Table XXI summarizes the selector outputs, and Fig. 14 shows the full grid-level sensitivity maps.

Across the tested reference-anchor grid  $D_0 \in \{7, \dots, 13\}$  and  $\eta_0 \in \{0.6, 0.9, \dots, 2.4\}$ , we find

$$f(\chi_{\mu\mu}^2 < 4) \in [0.0231, 0.2339], \quad \chi_{\min}^2 \in [2.28 \times 10^{-9}, 4.36 \times 10^{-2}]. \quad (\text{E12})$$

The baseline fixed-anchor value used in main figures is  $f(\chi_{\mu\mu}^2 < 4) = 0.1614$ . For reporting discipline, we keep this fixed anchor in headline plots and treat dynamic-anchor outputs as sensitivity diagnostics rather than post-hoc optimization.

## 7. Dephasing-Based First-Principles Candidate for $t_{\text{coh}}$

To close the finite-time module with the same action-derived spectral chain, we test the dephasing candidate

$$\Delta\omega_{12}(D) \equiv \omega_2(D) - \omega_1(D), \quad t_{\text{coh}}^{(\text{deph})}(D) \equiv \frac{\pi}{\Delta\omega_{12}(D)}, \quad (\text{E13})$$

TABLE XXII. 1D on-axis dephasing candidate for  $t_{\text{coh}}$  at  $D = \{6, 12, 18\}$  (coarse/mid/fine). Source CSV: `output/tcoh_fp_1d/tcoh_dephasing_D6-12-18.csv`.

| $D$ | level  | $\omega_1$ | $\omega_2$ | $\Delta\omega_{12}$     | $t_{\text{coh}}^{(\text{deph})}$ |
|-----|--------|------------|------------|-------------------------|----------------------------------|
| 6   | coarse | 0.720883   | 0.738989   | $1.8107 \times 10^{-2}$ | $1.7350 \times 10^2$             |
| 6   | mid    | 0.721345   | 0.739468   | $1.8123 \times 10^{-2}$ | $1.7334 \times 10^2$             |
| 6   | fine   | 0.721500   | 0.739629   | $1.8129 \times 10^{-2}$ | $1.7329 \times 10^2$             |
| 12  | coarse | 0.721781   | 0.722035   | $2.5438 \times 10^{-4}$ | $1.2350 \times 10^4$             |
| 12  | mid    | 0.722258   | 0.722513   | $2.5533 \times 10^{-4}$ | $1.2304 \times 10^4$             |
| 12  | fine   | 0.722418   | 0.722674   | $2.5565 \times 10^{-4}$ | $1.2289 \times 10^4$             |
| 18  | coarse | 0.719404   | 0.719408   | $3.6700 \times 10^{-6}$ | $8.5602 \times 10^5$             |
| 18  | mid    | 0.719885   | 0.719888   | $3.6943 \times 10^{-6}$ | $8.5039 \times 10^5$             |
| 18  | fine   | 0.720046   | 0.720050   | $3.7025 \times 10^{-6}$ | $8.4850 \times 10^5$             |

TABLE XXIII. Global-scan impact of the dephasing  $t_{\text{coh}}$  candidate (diagnostic only; not baseline).

| case                                | $f(\mathcal{R}_3 > 0.90)$ | $f(\mathcal{R}_3 > 0.95)$ | $f(\chi_{\mu\mu}^2 < 4)$ | best $\chi^2$         |
|-------------------------------------|---------------------------|---------------------------|--------------------------|-----------------------|
| constant $t_{\text{coh}} = 1$       | 0.983                     | 0.217                     | 0.1017                   | $5.47 \times 10^{-5}$ |
| $\pi/\Delta\omega_{12}(D)$          | 0.983                     | 0.217                     | 0.1017                   | $5.47 \times 10^{-5}$ |
| $\min(\pi/\Delta\omega_{12}, 10^4)$ | 0.983                     | 0.217                     | 0.1017                   | $5.47 \times 10^{-5}$ |

where  $(\omega_1, \omega_2)$  are extracted from the bound-state convention  $E_n = \omega_n^2 - m_0^2 < 0$  in the same 1D on-axis operator chain as Appendix B. The reproducible implementation is provided in the code directory (`extract_tcoh_dephasing_1d.py`).

For non-fine levels, the maximum relative deviation is

$$\max\left(\frac{|t_{\text{coh}} - t_{\text{coh}}^{(\text{fine})}|}{|t_{\text{coh}}^{(\text{fine})}|}\right) = 8.86 \times 10^{-3}, \quad (\text{E14})$$

which is numerically stable at the sub- $10^{-2}$  level for this benchmark set.

To assess map-level impact, we compared three cases on the same  $60 \times 60$  ( $D, \eta$ ) grid within the legacy constant- $A_\ell$  reference chain: constant  $t_{\text{coh}} = 1$ , raw dephasing profile from Eq. (E13), and a capped variant  $\min(t_{\text{coh}}^{(\text{deph})}, 10^4)$ . Results are summarized in Table XXIII (source CSV: `paper/tcoh_profile_impact.csv`).

The dephasing candidate leaves both  $\mathcal{R}_3$  and  $H \rightarrow \mu\mu$  summary fractions unchanged on this grid under the EFT/Wilson-matched diagnostic chain. Therefore, in this manuscript, Eq. (E13) is reported as a first-principles candidate benchmark only and is not propagated into the baseline reported figures.

TABLE XXIV. 1D on-axis benchmark for  $\eta$  prefactor candidates at  $D = \{6, 12, 18\}$  (coarse/mid/fine).  
Source CSV: `output/eta_fp_1d/eta_prefactor_D6-12-18.csv`.

| $D$ | level  | $\Delta E_{12}$         | $S_1$    | $A_{12}$ | $\eta_{\text{amp}}$ | $\eta_{\text{prob}}$ |
|-----|--------|-------------------------|----------|----------|---------------------|----------------------|
| 6   | coarse | $2.6433 \times 10^{-2}$ | 4.28295  | 1.91521  | 1.04764             | 1.09755              |
| 6   | mid    | $2.6475 \times 10^{-2}$ | 4.29274  | 1.93709  | 1.05923             | 1.12197              |
| 6   | fine   | $2.6489 \times 10^{-2}$ | 4.28158  | 1.91661  | 1.04787             | 1.09803              |
| 12  | coarse | $3.6727 \times 10^{-4}$ | 8.51270  | 1.82812  | 1.00000             | 1.00000              |
| 12  | mid    | $3.6889 \times 10^{-4}$ | 8.50865  | 1.82877  | 1.00000             | 1.00000              |
| 12  | fine   | $3.6944 \times 10^{-4}$ | 8.50734  | 1.82906  | 1.00000             | 1.00000              |
| 18  | coarse | $5.2804 \times 10^{-6}$ | 12.75507 | 1.82864  | 1.00028             | 1.00056              |
| 18  | mid    | $5.3189 \times 10^{-6}$ | 12.75845 | 1.84820  | 1.01063             | 1.02137              |
| 18  | fine   | $5.3320 \times 10^{-6}$ | 12.74576 | 1.82936  | 1.00017             | 1.00033              |

### 8. Splitting-Prefactor First-Principles Candidate for $\eta$

To test a geometry-closed replacement direction for the overlap amplitude in Eq. (24), we use the same action-derived quantities  $(\Delta E_{12}, S_1)$  and define the splitting prefactor

$$\Delta E_{12}(D) \approx A_{12}(D) e^{-S_1(D)}, \quad A_{12}(D) \equiv \Delta E_{12}(D) e^{S_1(D)}. \quad (\text{E15})$$

With a reference point  $D_{\text{ref}} = 12$ , we define two dimensionless candidates:

$$\eta_{\text{amp}}(D) \equiv \frac{A_{12}(D)}{A_{12}(D_{\text{ref}})}, \quad \eta_{\text{prob}}(D) \equiv \eta_{\text{amp}}(D)^2. \quad (\text{E16})$$

The reproducible script is `extract_eta_prefactor_1d.py` in the `code/` directory.

For non-fine levels in Table XXIV, we obtain

$$\max \left( \frac{|\eta_{\text{amp}} - \eta_{\text{amp}}^{(\text{fine})}|}{|\eta_{\text{amp}}^{(\text{fine})}|} \right) = 1.08 \times 10^{-2}, \quad \max \left( \frac{|\eta_{\text{prob}} - \eta_{\text{prob}}^{(\text{fine})}|}{|\eta_{\text{prob}}^{(\text{fine})}|} \right) = 2.18 \times 10^{-2}, \quad (\text{E17})$$

and the fine full-scan profile over  $D = 4, \dots, 20$  gives  $\eta_{\text{amp}} \in [0.9996, 1.1695]$  and  $\eta_{\text{prob}} \in [0.9992, 1.3677]$  (source CSV: `output/eta_fp_1d/eta_prefactor_D4-5-...-20.csv`).

To quantify map-level sensitivity, we compare a reference  $\eta_{\text{eff}} = \eta$  run (legacy constant- $A_\ell$  chain) against profile-scaled and fully-closed variants on the same  $60 \times 60$  grid. Results are summarized in Table XXV (source CSV: `paper/eta_profile_impact.csv`).

At current scan resolution, profile-scaled implementations remain close to the reference map-level summaries, while fully-closed implementations reduce  $f(\chi_{\mu\mu}^2 < 4)$  to 0.0833 in this diagnostic chain. Therefore, Eq. (E16) is reported as a first-principles candidate benchmark and is not propagated into baseline reported figures.

TABLE XXV. Global-scan impact of  $\eta$  prefactor candidates (diagnostic only; not baseline).

| case                                             | $f(\mathcal{R}_3 > 0.90)$ | $f(\mathcal{R}_3 > 0.95)$ | $f(\chi_{\mu\mu}^2 < 4)$ | best $\chi^2$         |
|--------------------------------------------------|---------------------------|---------------------------|--------------------------|-----------------------|
| reference $\eta_{\text{eff}} = \eta$             | 0.983                     | 0.217                     | 0.1017                   | $5.47 \times 10^{-5}$ |
| $\eta_{\text{eff}} = \eta \eta_{\text{amp}}(D)$  | 0.983                     | 0.217                     | 0.1022                   | $6.24 \times 10^{-5}$ |
| $\eta_{\text{eff}} = \eta \eta_{\text{prob}}(D)$ | 0.983                     | 0.217                     | 0.1019                   | $4.45 \times 10^{-6}$ |
| $\eta_{\text{eff}} = \eta_{\text{amp}}(D)$       | 0.983                     | 0.217                     | 0.0833                   | $6.34 \times 10^{-2}$ |
| $\eta_{\text{eff}} = \eta_{\text{prob}}(D)$      | 0.983                     | 0.217                     | 0.0833                   | $6.39 \times 10^{-2}$ |

### 9. Action-Derived First-Principles Candidate for Superradiant Channel Scaling

To test a geometry-closed replacement direction for the channel normalization in Eq. (32), we keep the same 1D action-derived chain and define channel-resolved barrier actions

$$U_\ell(z; D) \equiv U(z; D) + \frac{\ell(\ell+1)}{z^2 + \varepsilon^2}, \quad S_{N,\ell}(D) \equiv \int_{z-}^{z+} dz \sqrt{U_\ell(z; D) - E_N(D)}, \quad (\text{E18})$$

with  $E_N = \omega_N^2 - m_0^2$  in bound-state convention. We then define geometry-driven channel rates

$$\Gamma_{N,\ell}^{(\text{geo})}(D) \equiv \omega_N(D) e^{-2S_{N,\ell}(D)}, \quad (\text{E19})$$

and infer effective channel normalizations by matching to Eq. (32):

$$A_\ell^{(\text{fp})}(D; N) \equiv \frac{\Gamma_{N,\ell}^{(\text{geo})}(D)}{\omega_N(D) (\omega_N(D) M_*)^{4\ell+4}}. \quad (\text{E20})$$

Using  $(D_{\text{ref}}, N_{\text{ref}}) = (12, 2)$ , we define profile factors

$$\tilde{A}_\ell(D) \equiv \frac{A_\ell^{(\text{fp})}(D; N_{\text{ref}})}{A_\ell^{(\text{fp})}(D_{\text{ref}}; N_{\text{ref}})}. \quad (\text{E21})$$

The reproducible script is `extract_superrad_prefactor_1d.py` in the `code/` directory.

For non-fine rows in Table XXVI, the maximum relative deviations are

$$\max \left( \frac{|\tilde{A}_1 - \tilde{A}_1^{(\text{fine})}|}{|\tilde{A}_1^{(\text{fine})}|} \right) = 3.22 \times 10^{-2}, \quad \max \left( \frac{|\tilde{A}_2 - \tilde{A}_2^{(\text{fine})}|}{|\tilde{A}_2^{(\text{fine})}|} \right) = 3.78 \times 10^{-2}. \quad (\text{E22})$$

The fine full-scan profile over  $D = 4, \dots, 20$  yields  $\tilde{A}_1 \in [8.54 \times 10^{-6}, 4.10 \times 10^5]$  and  $\tilde{A}_2 \in [3.95 \times 10^{-6}, 4.92 \times 10^6]$  (source files in `output/superrad_fp_1d/`), indicating a very stiff  $D$ -dependence in this preliminary closure.

To quantify map-level sensitivity, we compare a legacy constant- $A_\ell$  reference  $(A_1, A_2) = (1, 1)$  against profile-scaled implementations on the same  $60 \times 60$  grid. Results are summarized in Table XXVII (source CSV: `paper/superrad_profile_impact.csv`).

At current scan resolution, replacing the legacy constant- $A_\ell$  reference by the action-derived profile induces non-negligible shifts in  $\mathcal{R}_3$  coverage, while the  $H \rightarrow \mu\mu$  acceptance area in this diagnostic chain remains unchanged. In the updated global scans of this manuscript, Eq. (E21) is used in baseline mode, and  $(A_1, A_2) = (1, 1)$  is retained only as a comparator.

TABLE XXVI. 1D on-axis benchmark for superradiant profile candidates at  $D = \{6, 12, 18\}$  (coarse/mid/fine). Source CSV: `output/superrad_fp_1d/superrad_prefactor_D6-12-18.csv`.

| $D$ | level  | $\omega_{N_{\text{ref}}}$ | $S_{N_{\text{ref}},1}$ | $S_{N_{\text{ref}},2}$ | $\tilde{A}_1$         | $\tilde{A}_2$        | $A_2^{(\text{fp})}/A_1^{(\text{fp})}$ |
|-----|--------|---------------------------|------------------------|------------------------|-----------------------|----------------------|---------------------------------------|
| 6   | coarse | 0.738989                  | 12.7478                | 20.7033                | $1.2936 \times 10^4$  | $5.5305 \times 10^4$ | $4.1247 \times 10^{-7}$               |
| 6   | mid    | 0.739468                  | 12.7644                | 20.7073                | $1.2452 \times 10^4$  | $5.4667 \times 10^4$ | $4.2192 \times 10^{-7}$               |
| 6   | fine   | 0.739629                  | 12.7711                | 20.7088                | $1.2652 \times 10^4$  | $5.6814 \times 10^4$ | $4.2589 \times 10^{-7}$               |
| 12  | coarse | 0.722035                  | 17.5746                | 26.3029                | 1.0000                | 1.0000               | $9.6480 \times 10^{-8}$               |
| 12  | mid    | 0.722513                  | 17.5720                | 26.3009                | 1.0000                | 1.0000               | $9.6103 \times 10^{-8}$               |
| 12  | fine   | 0.722674                  | 17.5866                | 26.3217                | 1.0000                | 1.0000               | $9.4844 \times 10^{-8}$               |
| 18  | coarse | 0.719408                  | 22.0000                | 31.0465                | $1.47 \times 10^{-4}$ | $7.9 \times 10^{-5}$ | $5.1810 \times 10^{-8}$               |
| 18  | mid    | 0.719888                  | 22.0060                | 31.0550                | $1.45 \times 10^{-4}$ | $7.8 \times 10^{-5}$ | $5.1411 \times 10^{-8}$               |
| 18  | fine   | 0.720050                  | 22.0043                | 31.0573                | $1.50 \times 10^{-4}$ | $8.0 \times 10^{-5}$ | $5.0954 \times 10^{-8}$               |

TABLE XXVII. Global-scan impact of superradiant profile candidates (diagnostic only; not baseline).

| case                                                     | $f(\mathcal{R}_3 > 0.90)$ | $f(\mathcal{R}_3 > 0.95)$ | $f(\chi_{\mu\mu}^2 < 4)$ | best $\chi^2$         |
|----------------------------------------------------------|---------------------------|---------------------------|--------------------------|-----------------------|
| legacy const $(A_1, A_2) = (1, 1)$                       | 0.9833                    | 0.217                     | 0.0833                   | $5.10 \times 10^{-2}$ |
| action-profile $(A_1, A_2) = (\tilde{A}_1, \tilde{A}_2)$ | 0.9269                    | 0.217                     | 0.0833                   | $5.10 \times 10^{-2}$ |
| $(A_1, A_2) = 0.5(\tilde{A}_1, \tilde{A}_2)$             | 0.9061                    | 0.217                     | 0.0833                   | $5.10 \times 10^{-2}$ |
| $(A_1, A_2) = 2(\tilde{A}_1, \tilde{A}_2)$               | 0.9469                    | 0.217                     | 0.0833                   | $5.10 \times 10^{-2}$ |

## Appendix F: Ansatz Ablation Study

To demonstrate that the three-generation structure is robust and not an artifact of a particular ansatz choice, we compare the baseline overlap-derived  $B_N(D)$  profile (this work) against the alternative “inverse Yukawa” ansatz explored in earlier versions.

### 1. Inverse Yukawa Ansatz (Ablation A)

The inverse Yukawa ansatz sets  $B_N^{(\text{inv})} \propto 1/\sum_{f \in \text{Gen } N} y_f$ , yielding:

$$B_1^{(\text{inv})} \approx 10^5, \quad B_2^{(\text{inv})} \approx 125, \quad B_3^{(\text{inv})} = 1. \quad (\text{F1})$$

This massive hierarchy ( $B_1 \gg B_2 \gg B_3$ ) directly encodes the answer by over-enhancing layer 1. While it can produce  $\mathcal{R}_3 > 95\%$ , the mechanism is less transparent: the visibility factor, rather than the entropic/kinetic competition, drives the result.



## 2. Comparison

- **Baseline (overlap-derived profile):**  $B_1(D) < B_2(D) < B_3$  in the scanned range, with smooth  $D$ -dependence from tracked microcanonical overlaps. The three-generation structure emerges from the  $g_N$ - $\Gamma_N$  competition modulated by these profile differences.
- **Ablation A (Inverse Yukawa):**  $B_1 \gg B_2 \gg B_3$ . Layer 1 is artificially boosted; the selection mechanism is less falsifiable.

We recommend the baseline as the primary closure due to its greater transparency and falsifiability.

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